

Measurements on the Cryogenic Hydrogen Beam

by

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I ensure that I have composed this paper independently and no source materials or aids other than those mentioned have been used, and all citations are labeled appropriately.

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1. Introduction and Motivation

1.1. Motivation

Purpose of this work is the development of a detector for atomic hydrogen and the subsequent characterisation of a cryogenic hydrogen beam. The work is performed in context of the Triton-Radius Experiment (T-REX) which aims at determining the charge radius of tritium.

The detector needs to fulfil the requirements of determining beam parameters such as flux, profile and divergence while not affecting the beam in order to provide the possibility of measuring during operation. Furthermore, a compact design is needed to find space in the small vacuum chamber.

This detector is not only applicable to T-REX, but can also be used wherever a beam from atomic hydrogen or atoms, which are equivalent in their behaviour, are to be measured in a vacuum environment.

The idea behind T-REX is to perform high-precision laser spectroscopy on tritium in order to determine the triton radius. The experiment aims at an increase of factor 300 according to the current precision of 5%, which could lead to a better understanding of the nuclear forces between the three nucleons.

In a microwave dissociation cavity, tritium molecules are split and then cooled down to 4.2 K with helium in a cryostat nozzle. The emitted atomic beam is purified in a magnetic velocity filter which leads only the lower end of the velocity distribution to a magneto optical trap (MOT). After having used lithium for further cooling, the lithium is removed and two-photon spectroscopy is performed. Because tritium is radioactive, hydrogen is used until the setup has been completed due to its similar characteristics.

The detector can be used behind the nozzle and the filter in order to determine its quality and analyse whether the flux to the MOT is high enough. As a result, the functionality of the components can be determined and its efficiency improved.

A new wire detector will be designed, which is an enhancement of a prototype which has formerly been built in the course of T-REX. In contrast to the current methods (quadrupole mass spectroscopy and pressure tube method), it combines the beneficial aspects of being selectively sensitive to atomic hydrogen, small enough to fit in the vacuum chamber and it does not severely affect the beam.

Following a design by Vassiliev et al., a thin wire can measure the recombination energy of the H-atoms. As its temperature increases, the resistance increases. By using a Wheatstone bridge, this resistance is transformed to a voltage which is

1. Introduction and Motivation

amplified and can be measured by a voltmeter.

However, there have been problems with changing resistance in vacuum, which caused the bridge to unbalance so much that the amplifier saturated. The prototype was not able to deliver a usable signal. Other attempts with only the wire and an ohmmeter worked for higher fluxes but did not provide the needed accuracy.

These problems are to be solved by developing a new detector, which uses an external unit for balancing and a better matched amplifier.

In the second part, measurements on the beam will be performed. The dependency of the flux on pressure and power are to be determined. Behind the nozzle, the beam will be characterised due to its profile, divergence and flux. The behaviour of the cryogenic beam will be observed, too.

1.2. Triton-Radius Experiment

The Triton-Radius Experiment (T-REX) is an experiment of the Pohl group at the Johannes Gutenberg-University Mainz. The goal of this experiment is to determine the radius of the tritium nucleus (triton) via high-precision laser spectroscopy of the 1S-2S transition in atomic tritium. As former scattering experiments have only reached a precision of 5 %, T-Rex aims at an improvement of two orders of magnitude. Of the lightest atoms, tritium is one candidate left where no high-precision measurements have been performed yet. However, due to its number of nucleons, it acts as the connecting particle between deuterium and helium. Having all the particles measured appropriately would allow further tests of three-nucleon forces and studies of isospin effects.[1]



The setup is shown in Figure 1.1: In a microwave dissociation cavity, tritium molecules are split into atoms. A tube guides the atoms to a helium-cooled nozzle (4.2 K), from where they, as a beam, can escape. Here, it is important not to give the atoms the opportunity to recombine. Therefore, materials with low adhesion coefficient for tritium, such as Teflon for hydrogen[2], need to be used. A quadrupole velocity filter[3] leads the slower part of the Maxwell-Boltzmann velocity distribution to a magneto-optical trap (MOT). Using lithium as buffer gas allows further cooling and trapping. Confined to a magnetic trap, 2-photon-spectroscopy is performed, which provides the information to compute the charge radius of tritium.[1]

1. Introduction and Motivation

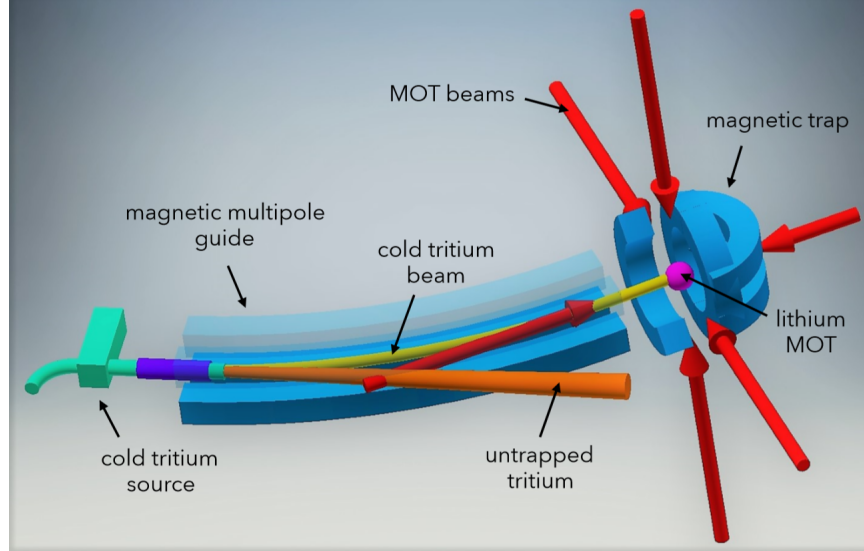


Figure 1.1.: Schematic T-REX setup[1]. The tritium atoms are cooled by a cryostat nozzle (4.2 K) and sent through a quadrupole velocity filter. Here, the slow part of the Maxwell-Boltzmann distribution is guided into a magneto-optical trap (MOT) filled with lithium atoms as buffer gas for further cooling. Afterwards, confined in a magnetic trap, two-photon laser spectroscopy is performed.

Since tritium is radioactive, the tests are performed with hydrogen instead, until setup has been finished. This works because they carry the same charge and the mass difference does not play a role for the general setup and working-principle of the experiment[4].

2. Design of an Atomic Hydrogen Detector

2.1. State of the Art

The first goal of this thesis is the development of a wire detector, as it combines several beneficial aspects against the common methods (mass spectroscopy, pressure tubes)[5]: It is selectively sensitive to atomic hydrogen, small enough to fit in the vacuum chamber and it does not severely affect the beam.

For T-REX, Schumacher[6] and Schwendler[7] have already built detectors of this type for measuring atomic hydrogen, using a thin tungsten wire as described in subsection 2.2.1.

Schumacher confirmed that a Wheatstone bridge is able to detect small changes of resistance and can be used for increasing the sensitivity to low resistances. His iteration one and two, however, did not work in the vacuum as the significant resistance change of the wire, caused by a change of pressure (dependency shown in section 3.1) led to an imbalance of the bridge and thus to saturation of the amplifiers used.

Schwendler was able to prove that the wire-model works with resistance probing of a single wire. He even was able to record a beam profile and estimate the atomic hydrogen flux. The detector used only consisted of a tungsten wire which was set in the beam. A laboratory multimeter measured the resistance with a probe current of 1 mA. Additionally, he showed that heating the wire to approximately 1000 °C and evaporating pollutions increases the signal (compare [8]), which will be taken advantage of in this thesis, too. However, this setup was not sensitive enough for measuring lower fluxes and therefore no measurements behind the cryostat have been performed.

2.2. The New Detector

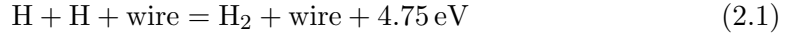
With respect to the findings of the previous experiments, a new detector can be designed, resolving the former issues. The numbering Schumacher introduced is continued and the new detector is called iteration 3.

As basis serves the concept of Schumacher, a Wheatstone bridge which is connected to an amplifier to increase the small differential voltage, so that a voltmeter can give out an appropriate result.

2. Design of an Atomic Hydrogen Detector

2.2.1. Wire-Model

The main idea for measuring atomic hydrogen is to measure its recombination energy when the atoms stick to a thin tungsten wire and are hit by another atom. When two H-atoms recombine to a H₂-molecule, the recombination energy of $E_{\text{rec}} = 4.75 \text{ eV} = 7.60 \times 10^{-19} \text{ J}$ [8] is set free. For comparison, the kinetic energy of a hydrogen atom at 100 m s^{-1} , is only $8 \times 10^{-24} \text{ J}$.



This energy heats the wire and thus increases its resistance, which can be measured. Having found the dependency of the change in resistance and the flux, one can determine the flux from the measurement. Thus, a heat-model of the wire as it can be found entirely in [9] is used:

The one-dimensional equation of the heat transfer

$$\lambda S \frac{d^2 T}{dx^2} = -F(x, T), \quad (2.2)$$

in which λ [W mK⁻¹] is the conductivity of the wire, S the cross-section of the wire, T the temperature and $F(x, T)$ [W m⁻¹] the density of the heat flow, describes the temperature distribution along the wire. As there are three contributions to the heat flow, $F(x, T)$ can be written as

$$F(x, T) = F_b(x) + F_{el}(T) - F_{rad}(T) \quad (2.3)$$

with the recombination heat flow density $F_b(x)$, which assumes a beam of diameter L_B and flux Q , hitting the wire of length L_w , diameter d and temperature T perpendicular with n atoms. It is supposed to be constant over the area from $-\frac{L_B}{2}$ to $\frac{L_B}{2}$. The heat flow density $F_{el}(T)$ through ohmic heating by the probe current I and the radiation contribution $F_{rad}(T)$ are given by

$$\begin{aligned} F_b(x) &= \frac{1}{2} \frac{n E_{\text{rec}}}{L_B} = \frac{2 Q E_{\text{rec}} d}{\pi L_B^2} =: C, \\ F_{el}(T) &= I^2 [R_0 + R_0 \alpha (T - T_0)], \\ F_{rad}(T) &= 4 \pi \sigma \epsilon d T_0^3 (T - T_0). \end{aligned} \quad (2.4)$$

Here, R_0 is the ohmic resistance, ϵ the emissivity of the wire and σ [J m⁻² s⁻¹ K⁻⁴] the Stefan-Boltzmann constant.

2. Design of an Atomic Hydrogen Detector

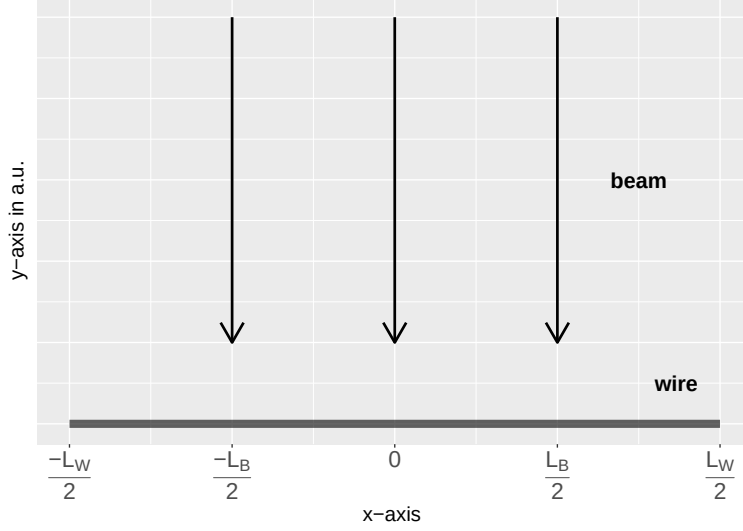


Figure 2.1.: Illustration of the wire with length L_W and the perpendicular beam of width L_B , taken from [6].

Inserting this into Equation 2.2 and assuming a fixed temperature T_0 at the edges of the wire as boundary conditions, leads to a solution describing the temperature distribution in three parts as

$$T(x) = T_0 + \frac{\beta_1}{A} + C_3 \cosh(\sqrt{A}x) \quad (2.5)$$

for the segment where the beam hits the wire and

$$\begin{aligned} T(x) &= T_0 + \frac{\beta_2}{A} + C_1 \cosh(\sqrt{A}x) + C_2 \sinh(\sqrt{A}x), \\ T(x) &= T_0 + \frac{\beta_2}{A} + C_1 \cosh(\sqrt{A}x) - C_2 \sinh(\sqrt{A}x) \end{aligned} \quad (2.6)$$

for the beam-free segments ($F_B = 0$) with the constants

2. Design of an Atomic Hydrogen Detector

$$\begin{aligned}
A &= \frac{4\pi\sigma\epsilon dT_0^3 - I^2 R_0 \alpha}{\lambda S}, \\
C_1 &= \frac{y_0 + (y_0 - y_1) \sinh(x_1) \sinh(x_0)}{\cosh(x_0)}, \\
C_2 &= (y_0 - y_1) \sinh(x_1), \\
C_3 &= \frac{y_0 + (y_0 - y_1) [\sinh(x_1) \sinh(x_0) - \cosh(x_1) \cosh(x_0)]}{\cosh(x_0)}, \\
x_0 &= \sqrt{A} \frac{L_w}{2}, & x_1 &= \sqrt{A} \frac{L_B}{2}, \\
y_0 &= -\frac{\beta_2}{A}, & y_1 &= -\frac{\beta_1}{A}, \\
\beta_1 &= \frac{C + I^2 R_0}{\lambda S}, & \beta_2 &= \frac{I^2 R_0}{\lambda S}.
\end{aligned} \tag{2.7}$$

Inserting the obtained temperature distribution 2.5 and 2.6 into the temperature dependency

$$R = \int_{-\frac{L_W}{2}}^{\frac{L_W}{2}} R_0 [1 + \alpha(T(x) - T_0)] dx \tag{2.8}$$

and then integrating over the length of the wire, leads to the total resistance of the wire

$$R_B = R_0 L_W + R_0 \alpha \left\{ -y_1 L_B - y_0 (L_W - L_B) + \frac{2(y_0 \sinh(x_0) - (y_0 - y_1) \sinh(x_1))}{\sqrt{A} \cosh(x_0)} \right\} \tag{2.9}$$

with the beam hitting the wire and

$$R_{B\text{off}} = R_0 L_W + R_0 \alpha \left\{ -y_0 L_W + \frac{2y_0 \sinh(x_0)}{\sqrt{A} \cosh(x_0)} \right\} \tag{2.10}$$

with the beam not hitting the wire.

The most accurate measurements can be achieved by choosing the thinnest possible wire with a preferably low probe current. This can easily be shown through numerical calculations[9]. An experimental proof of the advantages of low diameters has been adduced by Schwendler[7]. This fits the requirements of the experiment because a thin wire does not affect the beam noticeably.

2.2.2. Wheatstone Bridge

For low fluxes, the resistance of the wire only changes so little ($\text{m}\Omega$) that an external ohmmeter would not be able to detect any changes, whereas it would be possible to

2. Design of an Atomic Hydrogen Detector

detect small voltages down to μV . So, the resistance change is converted to a voltage using a Wheatstone bridge. This bridge allows very precise measurements of resistance by probing a differential voltage. Its wiring scheme is shown in Figure 2.2.

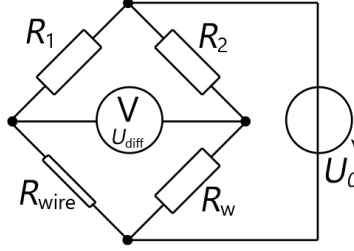


Figure 2.2.: Scheme of a Wheatstone bridge with wire R_{wire} and resistors R_w , R_1 , R_4 at the voltmeter V , a differential voltage U_{diff} is measured. U_0 is the supply voltage. (Image taken from [10] and edited.)

The differential voltage U_{diff} to be probed is zero if $R_{\text{wire}}R_2 = R_2R_1$. Ideally, all four resistors have the same resistance R which shall be achieved in the design, too. The resulting differential voltage then is $U_{\text{diff}} = 0$. If the change ΔR of resistor R_{wire} is to be measured and the other ones are known, the narrowed differential voltage U_{diff} , depending on the supply voltage U_0 yields to[11]

$$\frac{U_{\text{diff}}}{U_0} = \frac{1}{4} \frac{\Delta R}{R} \quad (2.11)$$

This circuit has proven appropriate for our needs and thus, will be used for the detector with the wire being R_1 in Equation 2.11.

2.2.3. Balancing the Bridge

As Schwendler has already shown, the bridge unbalances in vacuum due to indirect the pressure dependency of the wire's resistance[7], and this leads to the saturation of the amplifier. Thus, no reasonable data can be collected. To prevent this, the possibility to adjust the balance of the bridge is needed. In vacuum, the measured resistance of the wire increases by around 100 K to circa 370 K. As for measuring the resistance, a current is applied, the wire is heated. In vacuum, there is less convectional cooling which leads to a higher equilibrium temperature and thus, raises the temperature dependent resistance.

For a rough balancing, two parallel arms with trimming potentiometers R_{t3} as shown in Figure 2.4 are added. However, as we know the resistance of the $5\text{ }\mu\text{m}$ tungsten wire of length 5 cm is around 370 Ohms in vacuum, the bridge shall have this resistance on all four arms since this would be the optimal case (compare subsection 2.2.2).

For minimising the offset and maximising the precision, balancing to zero differential voltage would be a great advantage. Because the detector can not be accessed directly

2. Design of an Atomic Hydrogen Detector

when set up in the vacuum chamber, electronically controllable resistors are needed which are connected in series in the parallel bridge arm. Here, N-channel JFETs (junction gate field-effect transistors) are an appropriate solution, as they are conductive between drain and source until a voltage is applied to the gate. The electric field between gate and source (U_{GS}) widens the depletion layer between drain and source, and thus increases the resistance [12].

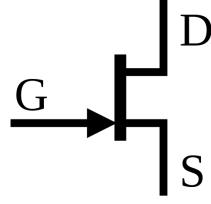


Figure 2.3.: JFET wiring symbol, taken from [13].

The needed trimming voltage U_{GS} can be applied by tuning the voltage divider displayed in Figure A.9, which is located in an accessible box outside of the vacuum chamber. Driven by 12 V, it allows to apply 0 to 6 V to the gate as it works like an adjustable bypass to the JFET. So, it is possible to trim very precisely in a range from $R_{JFET} = 0$ to 50 Ohms during operation. R_b has to be much larger than the total resistance of the bridge. Pulling down the supply voltage of 12 V, it limits the current such that the wire is less heated. Additionally, changes of the bridge's resistance do not intensely affect the current. That way, the measurement is more precise and stable. The specifications of the used JFET (J111) can be found in its data sheet[14].

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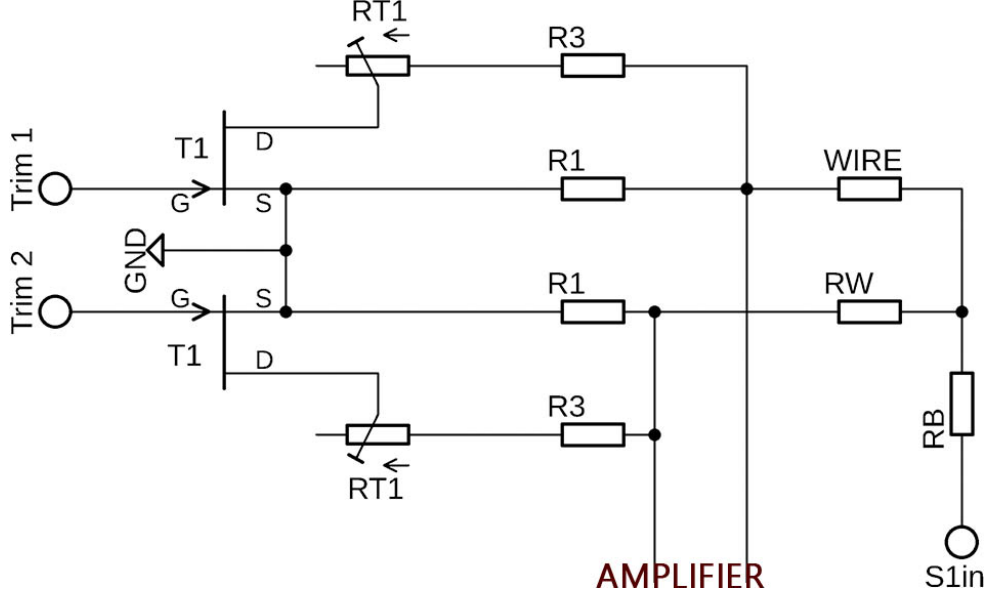


Figure 2.4.: Circuit diagram of the Wheatstone bridge as used in the detector. WIRE, R_1 , R_1 and R_w form the actual bridge. The parallel arms with R_3 , the JFET T_1 and trim resistor R_{T1} allow adjusting the effective resistance of R_1 . R_b pulls down the voltage to limit the current and improve stability. It is connected to the amplifier in Figure 2.5.

The total resistance of all four regular arms, whereat the arm with R_1 and its trimming component count as one, aims at being the same. However, trimming adds an unknown contribution. Since having all arms equal is the optimal case, this makes the detector less sensitive. By knowing the approximate resistance of each tuning device, it is possible to optimise the circuit anyway.

2.2.4. Amplifying the Signal

As signals down to $1\text{ }\mu\text{V}$ (at $\Delta R = 2.5\text{ m}\Omega$) are expected, which easily could be obscured by noise, an operational amplifier (opamp) is used, since a strong signal is qualitatively not affected. We aim for an output signal of at least a few millivolts for the weakest differential voltages.

Higher input voltages up to 40 mV (at $\Delta R = 100\text{ }\Omega$) should be measurable, too. Therefore, the gain has to be adjustable. The model chosen (LT1168[15]) provides a gain adjustable from 1 to 10^4 . Additionally, it will not be damaged by the voltage needed to heat out the wire (1.3 V), and thus suits the detector perfectly. The gain G can be chosen by joining up a resistor between two dedicated pins. Its value is determined by the formula $RG = 49.4\text{ k}\Omega/(G-1)$, which is taken from the data sheet[15]. As shown in Figure 2.5, using pin header JP_1 and a jumper, the gain

2. Design of an Atomic Hydrogen Detector

can easily be chosen between 1, 10, 10^2 , 10^3 and 10^4 by selecting the correspondent resistors.

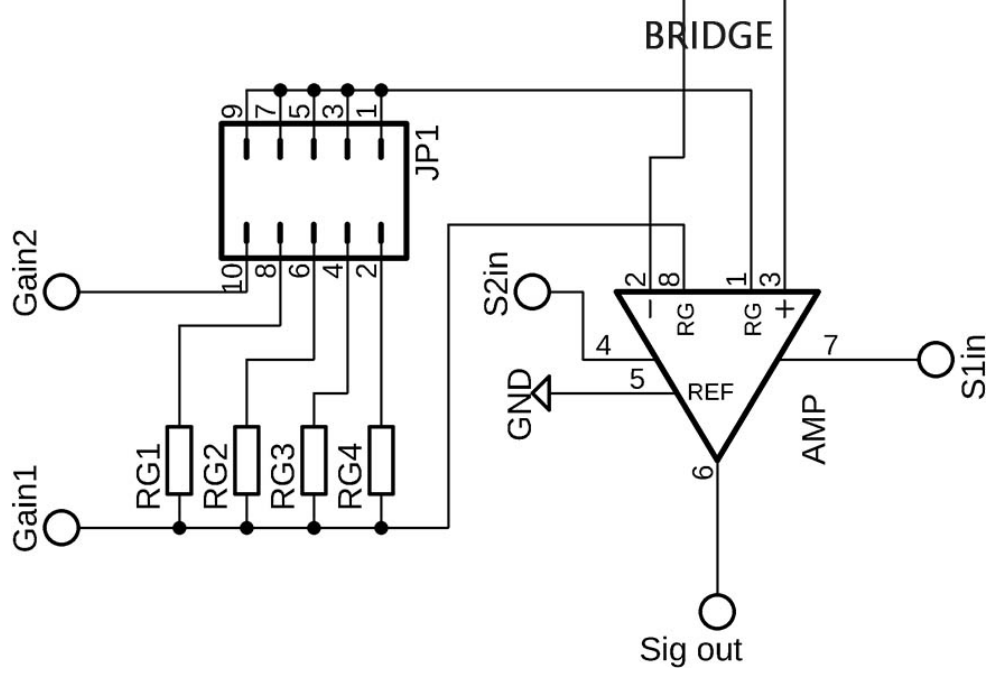


Figure 2.5.: Circuit diagram of the amplifier section as used in the detector, including the circuit for setting the gain. At JP1, the gain can be chosen by connecting one of the resistors with a jumper. The connectors Gain₁ and Gain₂ are used for adjusting the gain from the outside.

2.2.5. Final Setup

The final wiring scheme of the detector is depicted in Figure A.8. As voltage supply, a source is used which provides ± 12 V and GND. It has been amended by a standard-used low-pass filter setup (R_s , C_1 , C_2) to get rid of some noise. For each supply input (S_{1in} , S_{2in}), one separate filter is installed. The trimming unit and an additional gain switching unit are intended for an outside box for controlling the detector in-use.

While revision 2 used surface mounted electronic devices (SMD), a regular 'through hole' PCB is employed to avoid soldering problems that have occurred with SMD parts, since these are easy to damage and hardly seizable. Parallel to the tungsten wire connected to the Wheatstone bridge a simple single-wire[7] was installed, too. This allows comparison of the two methods. The detector is shown in Figure 2.6. On the right, the two wires are slightly visible. The Wheatstone bridge, except for the wire arms, is built symmetrically, as it could improve the balance. To get as few interferences as possible on the signal, there is a separate cable for it. The GND as reference is taken at the outside box.

2. Design of an Atomic Hydrogen Detector

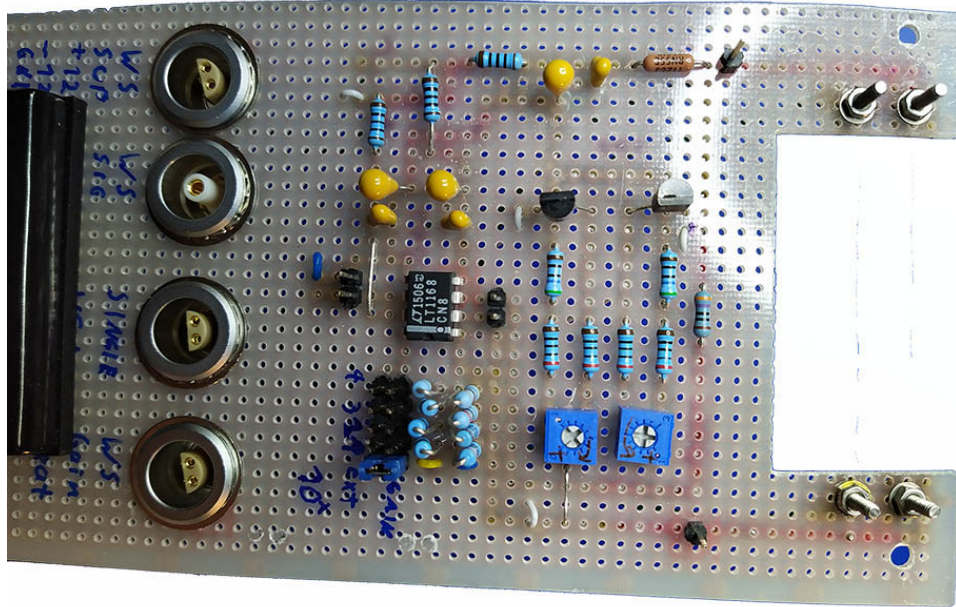


Figure 2.6.: Front view of the detector. On the right, two thin wires are recognisable. The right one is part of the Wheatstone detector, the left one of the single-wire detector.

It has proven to be the best if all connections to the outside are electrically shielded, especially the signal cables, as revealed in subsection 2.3.2. They lead to the also shielded and grounded outside trimming box (Figure 2.7). In this box, it is possible to set the gain if it is not set on the detector.

The shielding and the potentiometer R_{t3} as well as the outer gain switching unit have been added during the tests in section 2.3. However, the wiring schemes display the current device.

2. Design of an Atomic Hydrogen Detector

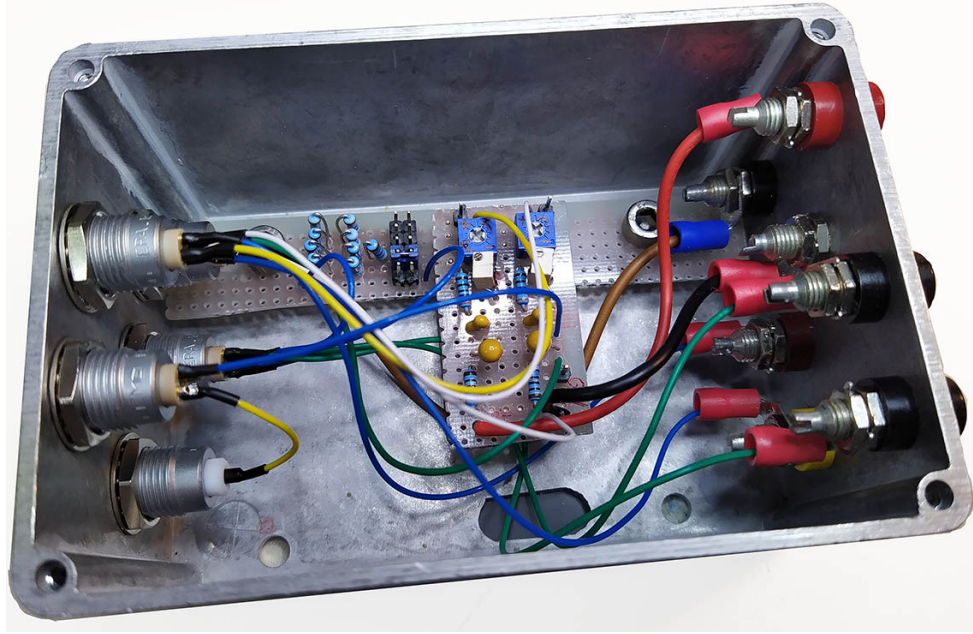


Figure 2.7.: The outside box. Used for balancing the bridge with the trim resistors (middle right) and setting the gain with a jumper at the related resistors (middle left). On the left, there are the connectors to the detector, on the right one can see the supply connectors.

2.3. Proof of Functionality

After the detector has been designed and built, its functionality needs to be proven. Additionally, a few improvements could be made.

2.3.1. Basic Functionality

To prove the basic functionality of the detector, it was first tested outside the vacuum chamber with a simple multimeter. For the first run, a $370\,\Omega$ metal-film resistor was used. Balancing the bridge with the trim potentiometers worked out as expected. With a gain of 10^2 a tuning to $40\,\text{mV}$, which corresponds to $400\,\mu\text{V}$ unamplified has been achieved. For further tuning, they were not fine enough. Using the JFETs for fine trimming, at a gain of 10^4 , a tuning to $30\,\text{mV}$, which corresponds to $3\,\mu\text{V}$ differential voltage at the bridge has been achieved. The stability of the signal was $\pm 2\,\text{mV}$ over a measured period of 20 seconds and $3\,\text{mV}$ over five minutes. These values have been taken during observation of the displayed voltage. Since only the basic functionality should be confirmed, no further data has been taken here.

2. Design of an Atomic Hydrogen Detector

Heating the resistor with the finger (while paying attention to not bypassing it) increased the differential voltage signal, a heat pistol provided the same result. It turned out that the $400\ \Omega$ trim potentiometers on the detector and the trimming unit are sensitive to touching it with a screwdriver. Thus, no exact data is expectable while they are modified. A $400\ \Omega$ fine trimming potentiometer R_{T3} (see Figure A.9) was added for more precise balancing.

Further tests were performed with a tungsten wire. Although the resistance of the wire at atmospheric pressure with $170\ \Omega$ differs from the $370\ \Omega$ in vacuum (10^{-5} mbar), it was possible to balance the bridge with the trim potentiometers on the detector due to their high range. At a gain of 10^2 , a balancing to $0.3\ \text{V}$, which corresponds to $U_{\text{diff}} = 3\ \text{mV}$ has been achieved. As the detector had been shielded in the vacuum chamber (but still with atmospheric pressure), the fluctuations, as could be observed on the voltmeter, decreased strongly. This shows that the wire is very sensitive to pickup noise. A further sensitivity which is considered to play a role is to air temperature fluctuations.

2.3.2. Functionality in Vacuum

At gain 10 a signal for both, normal pressure and vacuum, could be observed, the hydrogen beam switched off during the entire period. For lower pressure, the signal increased which implies that the resistivity of the wire increased like expected. Balancing the bridge was only partially possible due to the finite outside trim range compared to the high unbalanced signal. This issue could be solved by raw trimming the detector on-board to zero minus the offset in vacuum.

After this procedure and evacuating a second time, balancing the bridge from outside succeeded. At 4.6×10^{-5} mbar, gain 100, a balancing range of $400\ \text{mV}$ ($\hat{=} U_{\text{diff}} = 1\ \text{mV}$) was achievable. The fine tuning was limited to $0.1\ \text{mV}$ ($\hat{=} U_{\text{diff}} = 1\ \mu\text{V}$) by the noise. This proves the given concept to be an appropriate solution to the detuning problems of the previous revisions.

2.3.3. Tests with Hydrogen

When positioning the wire close (1-2 mm) to the nozzle and starting the molecular hydrogen beam, a diminution of the signal could be seen. The molecular hydrogen is assumed to cool the wire, and thus decrease its resistance (see subsection 3.2.1). When starting the dissociation (described in subsection 3.1.1), the signal increased strongly. The dependence of the signal on pressure and dissociation power are accounted in section 3.1 and prove the detector to be working.

One improvement that reduced the noise by factor ten was shielding the signal banana cables, which are needed for the last 30 cm to the voltmeter.

3. Measurements

3.1. Investigating the Dissociation

An important part of the Triton-Radius Experiment is the production of atomic hydrogen in the dissociation area. As the atomic hydrogen is generated there, it is the first component to be optimised in order to guarantee a maximum H-atom flow towards the magnetic trap. These measurements are made without the copper nozzle.

3.1.1. The Dissociation

The microwave dissociation generates atomic hydrogen by dissociating hydrogen molecules in a microwave dissociation cavity. As depicted in Figure 3.1, it is built from a glass tube of 6 mm diameter, which is connected to a constant-pressure hydrogen supply by a needle valve on the one side and to a vacuum chamber by a Teflon hose on the other side. Teflon fits well to prevent the atoms from recombining due to its low sticking coefficient for atomic hydrogen. Thus, the atom flow can be guided over longer distances without any greater losses[2]. For tritium, an other material has to be chosen which tolerates the radioactivity. An impedance of diameter (0.5 ± 0.2) mm causes a pressure difference between vacuum chamber and dissociation area, which can be adjusted with the valve so that the resulting equilibrium state between inlet and outlet leads to a constant pressure, which is optimal at 2 to 4 mbar. It is measured and displayed by a vacuum meter.

In the microwave cavity, which is driven at 2400 to 2500 Hz, a nearby initial electric discharge sets electrons free. These are accelerated by the standing microwave and, when hitting hydrogen atoms, dissociate them. The atoms then can escape through the impedance into the vacuum chamber.

High density of H-atoms and microwave power increase the dissociation rate. However, while a high dissociation rate is needed, power and pressure should be low as they affect the experiment: because of the pressure in the dissociation area, the pressure in the vacuum chamber, which should be as low as possible, increases. The dissociation is heating up the gas to temperatures high enough to burn the glass. Therefore cooling is applied. However, driving the dissociation at high power ($P > 50$ W) results in rising temperatures of the gas yet again, which resumes the aforementioned issue of burning the glass.

3. Measurements

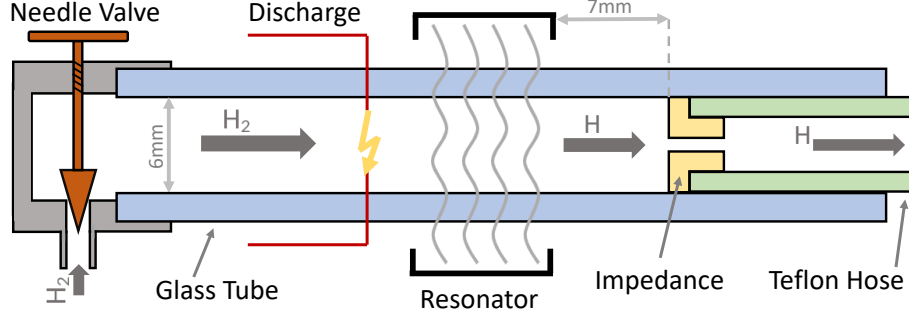


Figure 3.1.: Sketch of the microwave dissociation. The hydrogen flow from a bottle into the glass tube is regulated by a needle valve. In the resonator, the molecules are dissociated by free electrons which are accelerated by a microwave. The discharge once supplies initial electrons to start the dissociation. Hydrogen atoms can escape through the impedance, guided by a Teflon hose, into the vacuum chamber. The pressure inside the dissociation area depends on the flow, which is regulated by the needle valve.

3.1.2. Dissociation Power and Pressure Dependency

A short look is taken at how pressure and power affect the beam intensity. The power mentioned matches the delivered power ($P = P_{\text{in}} - P_{\text{out}}$) and the pressure is the one in the dissociation area which differs from the one in the vacuum chamber ($\sim 3.7 \text{ mbar}$ to $\sim 1 \times 10^{-5} \text{ mbar}$).

It needs to be mentioned that the microwaves in the dissociation area without any beam do not affect the measurement. For demonstration see Figure A.1, where no effect smaller than the normal fluctuations is visible. An other effect is the improvement of the dissociation over the first minutes after ignition. This effect is visible in Figure A.3.

In Figure 3.2, the atomic hydrogen signal is plotted against the microwave power. The signal increases with the power. Regarding subsection 2.2.1 and Equation 2.11, this means an increase of H_1 flux, and thus confirms our expectations. However, the slope decreases and so it does not pay off to raise the power to more than 40 W. This non-linear behaviour results from the velocity distribution of the electrons. Increasing the power makes more electrons exceed the minimum energy that is needed to split the hydrogen molecules until saturation, where nearly all molecules are dissociated. Thus, the hydrogen flux which increases the detector signal can not ascend further. As higher power would damage the dissociation area (burning the glass tube), showing the complete saturation at higher power is not possible. Like in [2], the saturation should be reached at 50 W to 60 W. It could also be noticed that the pressure in

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the dissociation area slightly varies with the microwave power. Higher power leads to higher pressure, while the needle valve has not been manipulated. The dissociation heats the gas and splitting the atoms leads to an increase of particles. Regarding to the ideal gas equation the pressure raises. So, this behaviour meets the expectation.

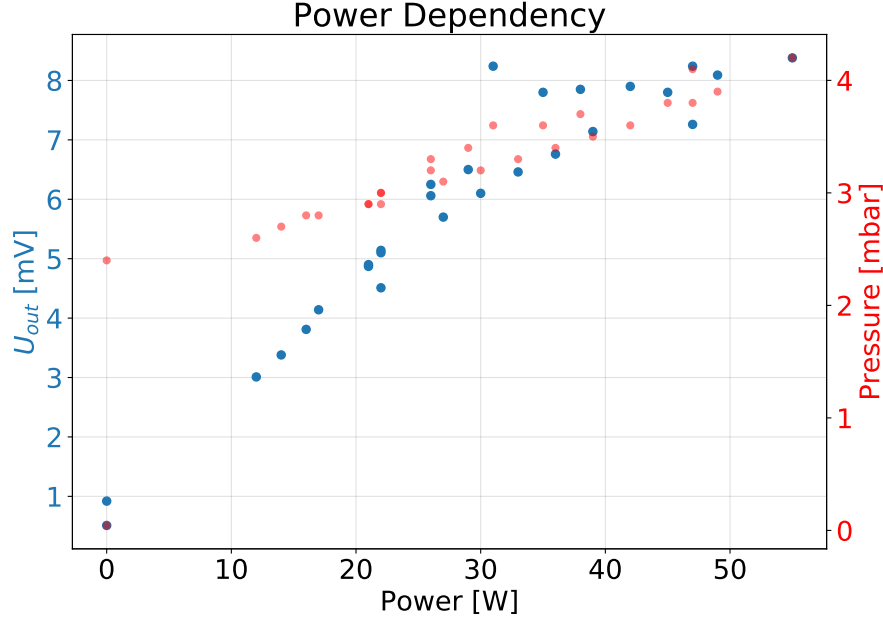


Figure 3.2.: Signal dependency of the microwave power with hydrogen beam (blue). The signal increases with the power and converges against saturation. An offset results from a slight unbalancing of the Wheatstone bridge. Also shown is the pressure (red) which increases with the power.

In Figure 3.2 which depicts a pressure dependency measurement with constant microwave power, the increasing signal is recognisable. As with power, the dissociation can not increase arbitrarily by simply raising the pressure. A possible explanation could be the mean free path which decreases with higher pressure. Fewer electrons reach the necessary kinetic energy that is needed for dissociation. Simultaneously, the number of potential targets increases, which raises the dissociation rate in the first place.

At around 1 mbar and at 3.8 mbar, the orange values do not fit the profile. Chronologically, the highest value was measured first, then the other ones with decreasing pressure. After the dissociation had turned off at 0.4 mbar due to a lack of molecules, it had to be reignited at a higher pressure. The values measured afterwards represent the main curve. This measurement was performed directly after analysing the power dependency. The high power used could have affected the dissociation longer as it

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was applied. In agreement to this, the red value is higher, as it was measured directly after probing for 5 mbar.

The following measurements are taken with a pressure of 2 mbar to 4 mbar, as the signal to pressure ratio is best in this range. Higher pressure causes more hydrogen to flow into the vacuum chamber and pollute the vacuum.

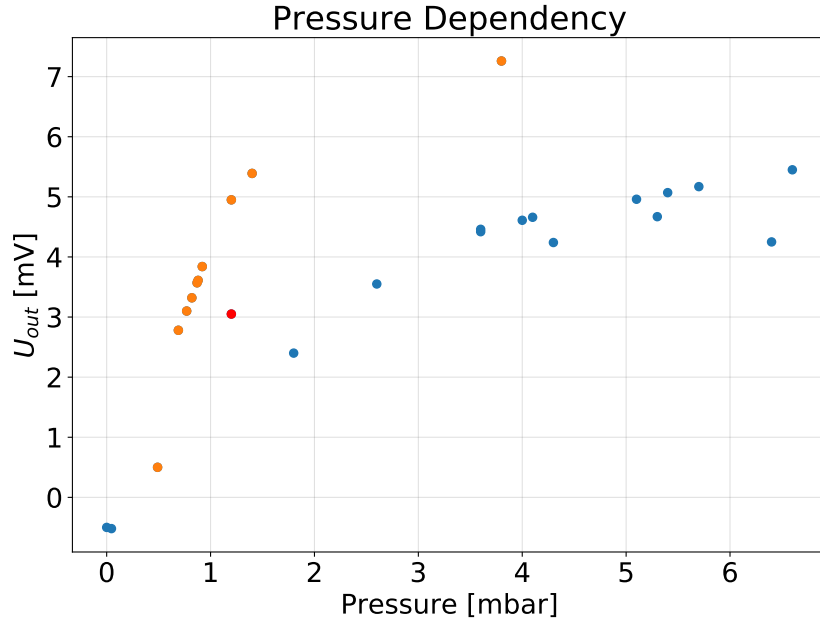


Figure 3.3.: Dissociation area pressure dependency of the signal: performed at 1 mm distance to the Teflon hose with 47 W microwave power and no amplification. The measurement was performed on the atomic hydrogen beam. A saturation behaviour can be observed. The values which do not fit the curve are coloured orange and red. Their irregularity is explained in the paragraph above.

3.2. Measurements behind the Cryostat

The next set of measurements was performed behind the cryostat, firstly with room temperature and then with helium cooling at 4.2 K. The detector is mounted on a linear stage, which can translate perpendicularly or parallel to the beam, depending on its orientation. A linear table allows vertical arrangement (Figure 3.4). As copper allows the atoms to stick and recombine on its surface, the signal is expected to be lower than without the nozzle.

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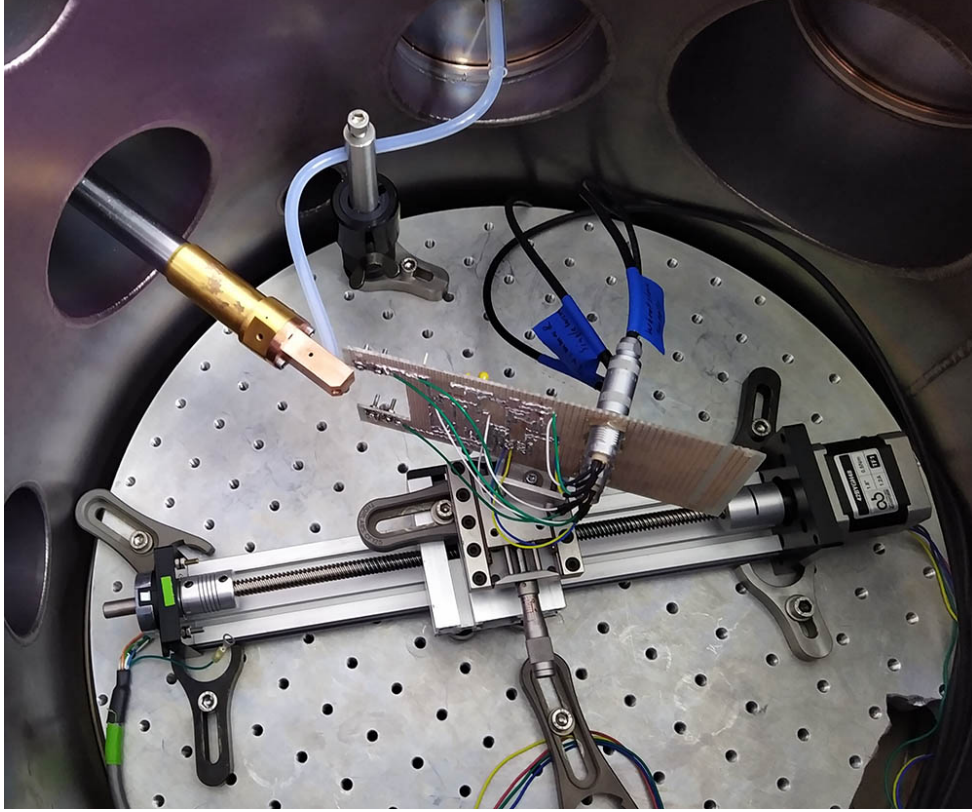


Figure 3.4.: Top view of the open vacuum chamber. Through the white Teflon hose, hydrogen is led to the copper nozzle of the cryostat. The detector is mounted on an electronic linear stage for perpendicular movement to the beam and a micrometer screw allows adjustment of the distance to the nozzle.

3.2.1. Probing the Beam at Room Temperature

Before measurements on the cryogenic beam are performed, the profile of the beam is probed at room temperature. To do so, the detector is attached to a linear stage which can move the detector perpendicularly to the beam. In small steps, the signal of the detector is measured and assigned to the position. The voltmeter takes mean values from ten measurements in one second per point. Then, multiple runs are averaged. The standard deviation serves as error value.

The beam profile of atomic hydrogen is depicted in Figure 3.5 together with the corresponding one of H_2 . In the caption, additional measurands are provided.

As stated in subsection 2.2.1, the recombination energy of H_1 heats the wire and increases its resistance which is proportional to the signal, according to Equation 2.11. For H_2 , which deposits no energy on the wire, a negative signal is detected. In vacuum,

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there is no convective cooling of the wire. However, the hydrogen can absorb heat and cool the wire. Its resistance decreases and a drop of the signal can be observed within the beam. This effect should be equivalent for atomic hydrogen. However, the heating by recombination dominates. As an approximation which neglects the varying energy transfer for different temperature differences the difference of the two signals has to be regarded. With and without dissociation, an offset results from the bridge unbalancing. Like in Schwendler's thesis, the beam profiles are Gaussian-shaped.

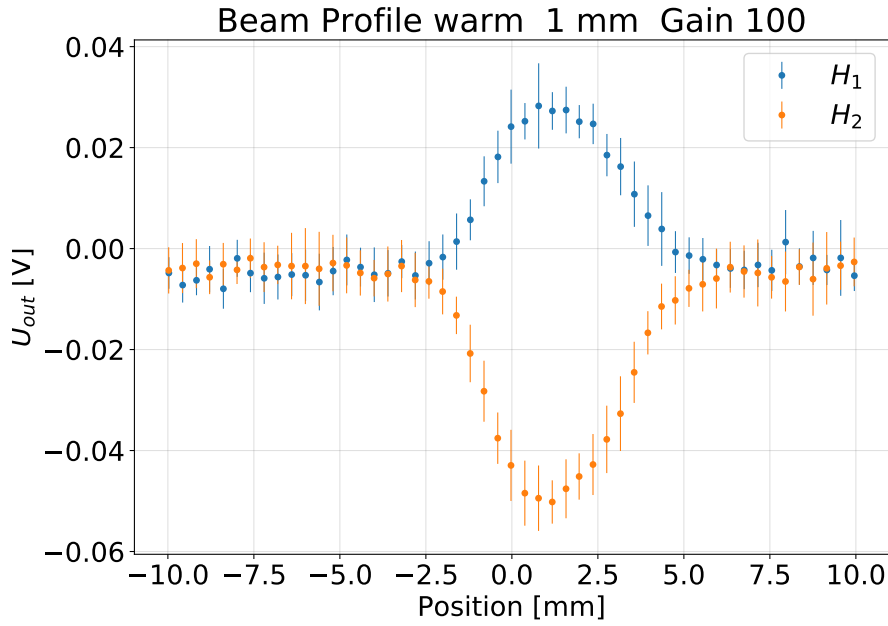


Figure 3.5.: Beam profile of H_1 and H_2 at 1 mm distance to the nozzle and 293 K. For H_1 , at a pressure of 2.5 mbar, 40 W microwave power are applied. The signal is amplified with a gain of 10^2 . For each value, nine points are averaged.

3.2.2. Probing the Cryogenic Beam

While cooling the cryostat down with liquid helium, measurements were taken 1 mm from the nozzle the same way as in subsection 3.2.1 but with fixed position. There are no errorbars provided since the averaging voltmeter does not provide the standard deviation. Simultaneously, the temperature was measured. Plotting temperature and the detector signal against the time in Figure 3.6, some noticeable effects are recognisable.

The first time, only an offset was detected since there was no beam. The cryostat is at room temperature ((293 ± 1) K). At the first vertical mark, shortly after the cooling

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had been started, the H_1 beam was activated. A huge increase of the signal shows the measured atomic hydrogen. Although the hydrogen was cooled, which opposes the deposited recombination heat on the wire, the signal increased over the first six minutes. This is due to an increase of the dissociation quality, which is visible in Figure A.3 and could be observed during other measurements as well.

While the temperature was falling with approximately 14 K min^{-1} , the detector signal reached its maximum and decreased to a minimum slightly above the offset as the temperature reached its minimum of 4.2 K. The strong decrease could be caused by the hydrogen cooling the wire, and thus lowering its resistance as well as a higher recombination rate on cold copper.

An other effect is the raise of the signal a short period after the cryostat had reached 4.2 K. The right marker shows the freezing temperature of hydrogen at normal pressure (14 K). In vacuum, it should be lower but still above the cooling temperature[2]. As an ice layer of hydrogen covered the copper, the signal could increase due to less recombination on the copper surface. This effect is also well recognisable in Figure A.2. A negative effect of freezing hydrogen is, that after around one hour, the nozzle freezes up and the beam is heavily disrupted. The pressure inside the dissociation area increases while it remains stable in the chamber and the detector signal decreases. This affected all measurements with helium cooling.

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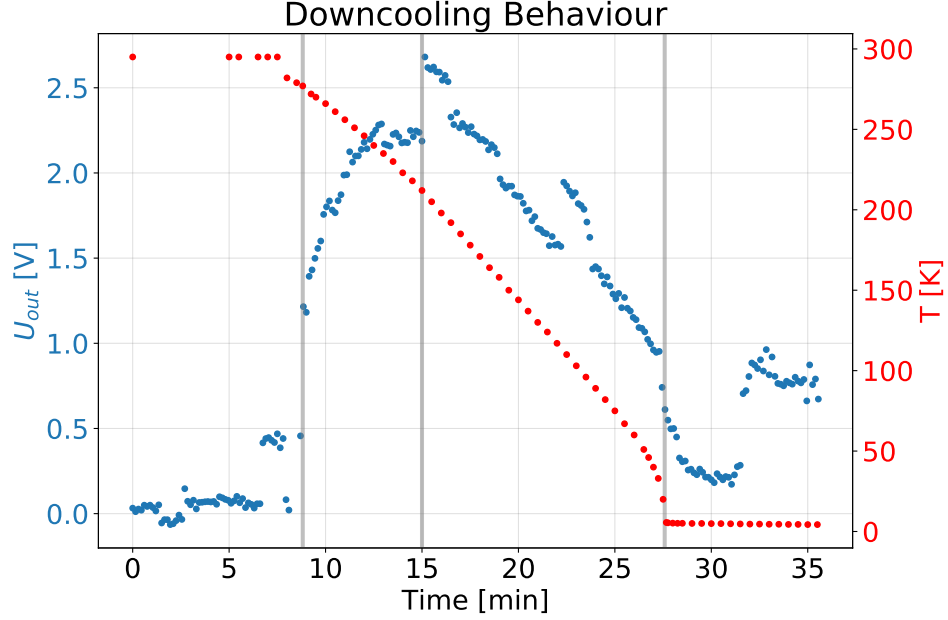


Figure 3.6.: Cooling down behaviour of the beam at 1 mm distance to the nozzle with 3.9 mbar in the dissociation area and 24 W microwave power. The signal is amplified with a gain of 10^3 . At the first grey mark, hydrogen was injected and the dissociation ignited. In the middle, the flux of atomic hydrogen reached its maximum. The right mark shows the temperature reaching the freezing point of hydrogen and then an increase of the signal.

The profile 1 mm behind the cryostat at 4.2 K was measured using the same methods as in subsection 3.2.1. The atomic hydrogen profile is clearly visible. Whereas the profile of the warm beam is shaped Gaussian-like, this profile looks more like a triangle. Frozen hydrogen could change the properties of the nozzle and affect the beam shape. For H_2 , only two runs (2 data points per position) could be performed because after that, the freezing nozzle disrupted the beam. As the negative signal of warm H_2 was about as strong as the one for warm H_1 , here it has only around two thirds the strength. An advanced freezing of the nozzle caused by a longer cooling time and enforced by a lack of recombination energy which heats the nozzle could be the reason. The wire sometimes touched the nozzle which led to grounding and the saturation of the amplifier. Also a temperature increase of the cryostat by around 20 K was visible. When the wire moved away from the nozzle, signal and temperature normalised.

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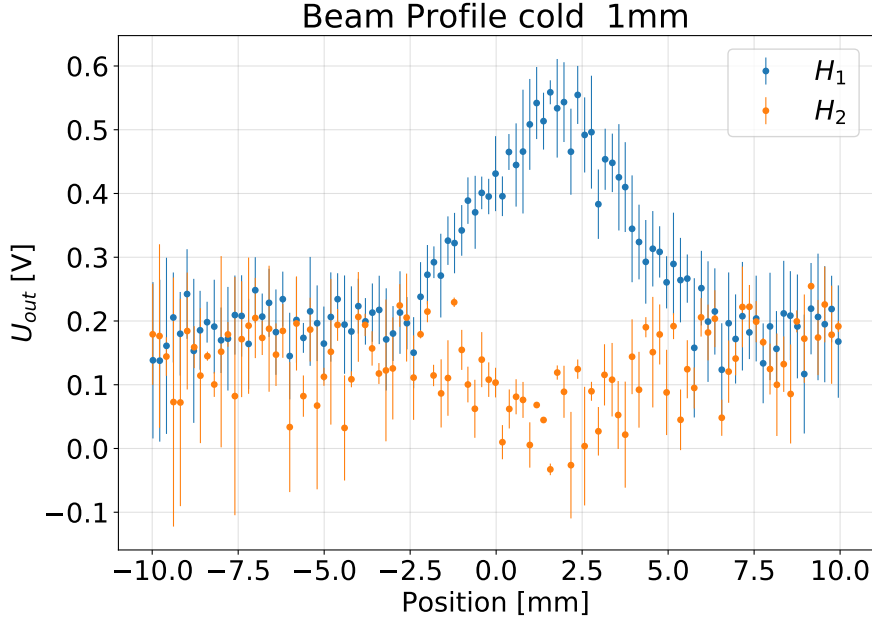


Figure 3.7.: Beam profile of H_1 and H_2 at 1 mm distance to the nozzle and 4.2 K. For H_1 , at a pressure of 3.7 mbar, 24 W microwave power are applied. H_2 is probed at a pressure of 2.6 mbar. The signal is amplified with a gain of 10^3 . For the H_1 values, six points are averaged. For H_2 , only two points, as further values have been affected too much by the freezing of the nozzle.

A similar measurement was performed at three millimetres distance to the nozzle. For this, the chamber had to be opened to move the detector with the micrometer screw. The measurement is depicted in Figure 3.8. Many values had to be dropped as the signal decreased to the negative saturation voltage of the amplifier or did not show the beam profile as the nozzle froze up. Only two data points could be averaged for H_1 , at -2.5 mm to 0 mm, only one value was in a realistic range. From the measurement for molecular hydrogen, which had been performed after the one of H_1 , only one run could be used. The data points are scattered very far but still show the negative profile caused by cooling of the wire.

The shape of the beam is affected more strongly than in Figure 3.7. Asymmetric deposit of hydrogen on the nozzle can cause such variations. The signal strength decreased as the width of the profile increased. A bigger beam cross-section at higher distance makes fewer particles collide with the beam and deposit recombination heat. How the signal is depending on the distance is analysed in subsection 3.2.3.

3. Measurements

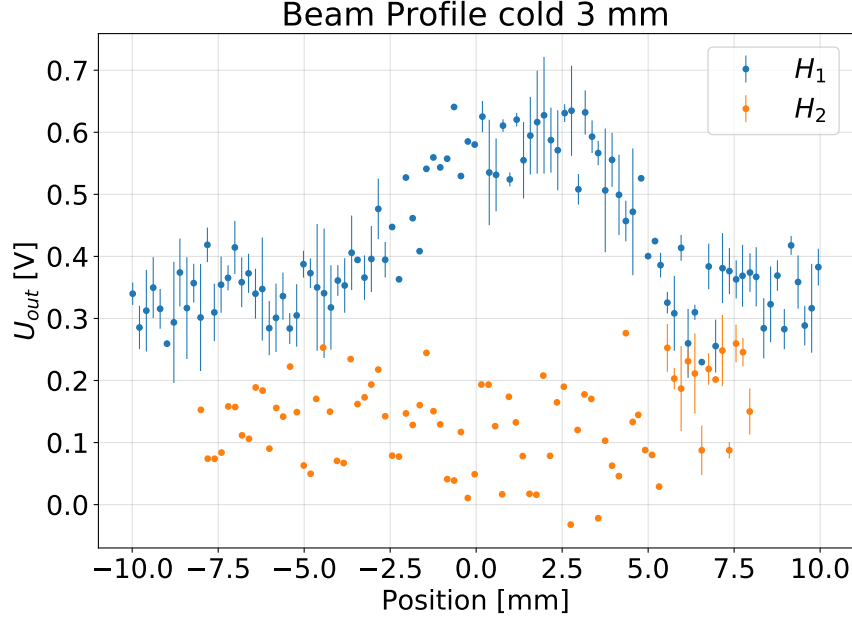


Figure 3.8.: Beam profile of H_1 and H_2 at 3 mm distance to the nozzle and 4.2 K. For H_1 , at a pressure of 3.7 mbar, 24 W microwave power are applied. H_2 is probed at a pressure of 3.0 mbar. The signal is amplified with a gain of 10^3 . For the H_1 values, mostly three points are averaged. For H_2 , only one run gave data which was not the saturation voltage of the amplifier. Grounding of the wire at the nozzle could have caused this.

3.2.3. Distance Measurement and Beam Divergence

Placing the linear stage parallel to the beam allows probing the beam depending on its distance. Like this, a measurement was performed at a cryostat temperature of 4.2 K, dissociation pressure of 3.6 mbar and power of 24 W. The signal is expected to be proportional to the inverse beam cross section and thus, to x^{-2} , where x is the distance from the nozzle, assuming a cone-shaped beam.

3. Measurements

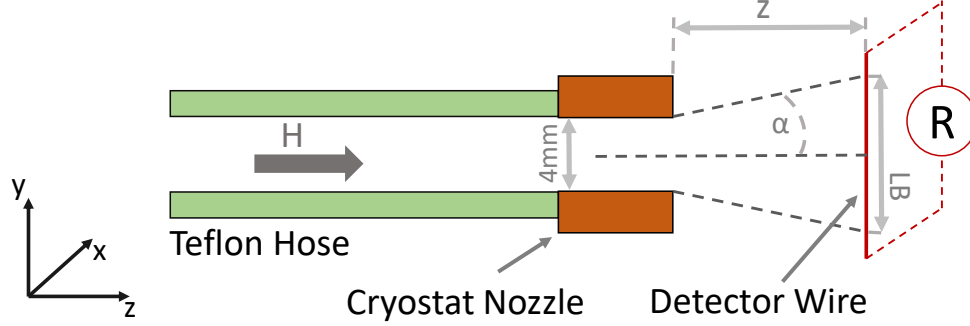


Figure 3.9.: Sketch of the beam from the cryostat. After being emitted by the cryostat nozzle, the beam broadens by angle α over the distance z to the wire. The wire can be translated in x or z direction with a linear stage.

The function

$$U(x) = \frac{a}{x^2} + \frac{b}{x} + c \quad (3.1)$$

was adjusted to the data shown in Figure 3.10 and follows the declining signal. The data is taken as described in subsection 3.2.1. So, those points without error bars were averaged of a period over one second, but no error could have been calculated as the freezing nozzle corrupted the measurement. From a certain time on, the signal decreased to the offset. In the first run, after the signal had reached a relative stable value at 4.2 K, driving away from the nozzle, the signal fluctuated over around 0.17 V. These fluctuations are slightly bigger than those of the beam profiles measured before. In the associated cool down measurement shown in Figure 3.6, the signal of around 1.7 V is fluctuating as well. This can be a characteristic of the cool hydrogen beam as it is not completely stable. But changes on the detector which occurred while relocating the stage for this measurement can not be excluded, too.

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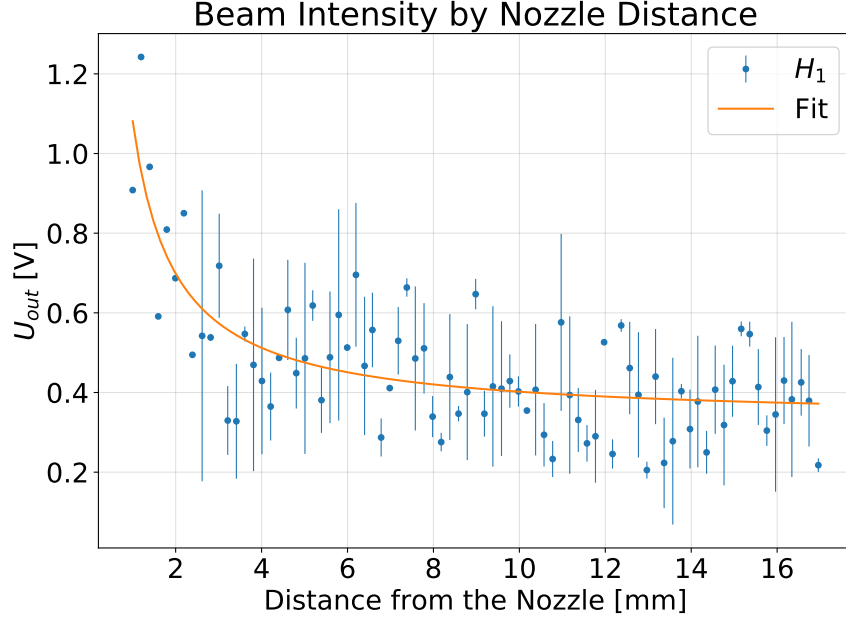


Figure 3.10.: Signal of cryogenic H_1 (4.2K) depending on the distance to the nozzle with 3.9 mbar in the dissociation area and 24 W microwave power. The signal behaves like Equation 3.1 with parameters $a = (0.17 \pm 0.27)$ V mm, $b = (0.58 \pm 0.23)$ V mm, $c = (0.34 \pm 0.03)$ V. For some points, error values can not be provided.

After the nozzle, the beam has to travel a way to the magnetic bottle. It would be optimal to carry as much hydrogen as possible to this trap and therefore only have a slight divergence. The divergence is also needed to calculate the atomic hydrogen flux. Depicted in Figure 3.9 is the nozzle of radius 2 mm which emits the beam to the wire along the z-axis. The beam diverges with an angle α_s .

Assuming the beams to have a Gaussian profile, an according fit of the function

$$\Delta U_{out} = a \exp \left[\frac{-(x - \mu)^2}{2\sigma^2} \right] + b \quad (3.2)$$

gives its width σ . From the profiles depicted in Figure 3.7 and Figure 3.8, measured at 1 mm and 3 mm behind the cold nozzle, the beam divergence can be calculated geometrically. The broadening corresponds to the linear inclination of σ from the fits with

$$\alpha_s = \arctan \left(\frac{\sigma_2 - \sigma_1}{z_2 - z_1} \right). \quad (3.3)$$

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The angle of enlarging that was found from the fits, specified in Table 3.1 and depicted in Figure 3.11, is $(16 \pm 5)^\circ$. Due to a lack of usable data points, the 3 mm fit has bigger errors. This increases the relative error on the divergence angle strongly due to error propagation. However, the angle corresponds to the measurements of Schwendler. He came up with an angle of $(15.9 \pm 0.3)^\circ$. Noticeable is that his measurement was performed without the cold nozzle but just a Teflon hose at room temperature. The determined value is used for calculating the atomic hydrogen flux.

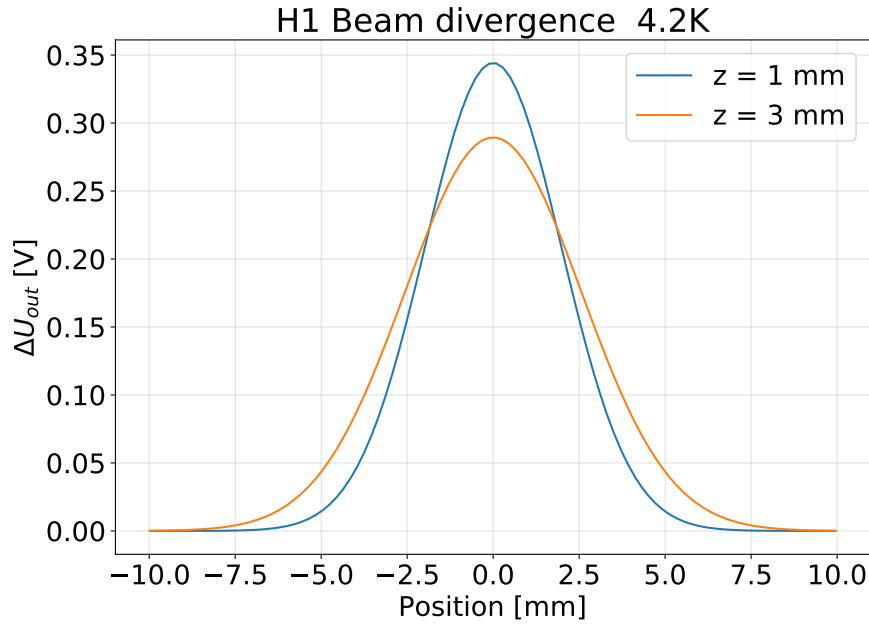


Figure 3.11.: Gaussian fits of atomic hydrogen beam profiles at 4.2 K and distances of 1 mm and 3 mm to the nozzle. The associated beam profiles are plotted in Figure 3.7 and Figure 3.8.

z-distance	1 mm	3 mm
a	$(0.344 \pm 0.008) \text{ V}$	$(0.289 \pm 0.014) \text{ V}$
b	$(0.193 \pm 0.005) \text{ V}$	$(0.329 \pm 0.008) \text{ V}$
μ	$(1.66 \pm 0.05) \text{ mm}$	$(1.27 \pm 0.13) \text{ mm}$
σ	$(1.99 \pm 0.06) \text{ mm}$	$(2.57 \pm 0.17) \text{ mm}$

Table 3.1.: Fit parameters corresponding to Equation 3.2 specifying the two Gaussian fits in Figure 3.11.

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3.2.4. Determining the atomic Hydrogen Flux

To evaluate the flux, we need to perform a few calculations. With the divergence angle α_s , according to Figure 3.9, the part of the wire inside the beam is

$$L_B = 2\sqrt{(z \tan(\alpha_s) + r)^2 - x^2} \quad (3.4)$$

with z being the distance of the wire to the nozzle, x the perpendicular distance to the beam center and r the nozzle radius of 2 mm. For calculating the resistance from the measured voltage, current I_0 through the wire and the voltage U_B applied to the bridge are calculated. The resistance of one of the trimmable arms of the Wheatstone bridge which is shown in Figure 2.2 is

$$R_{\text{arm}} = \left(\frac{1}{R_1} + \frac{1}{R_3 + R_{t1} + R_t} \right)^{-1} \quad (3.5)$$

Two parallel arms and the wire and its parallel R_w build the bridge with resistance

$$R_{\text{bridge}} = \left(\frac{1}{R_{\text{arm}} + R_w} + \frac{1}{R_{\text{arm}} + R_{\text{wire}}} \right)^{-1} \quad (3.6)$$

At the bridge, a voltage of

$$U_{\text{bridge}} = U_0 \left(1 - \frac{R_b + R_s}{R_b + R_s + R_{\text{bridge}}} \right) \quad (3.7)$$

is applied. A current of

$$I_w = U_{\text{bridge}}(R_{\text{arm}} + R_{\text{wire}}) \quad (3.8)$$

flows through the wire. The resistance of the wire can be calculated from the measured voltage U_{sig} by using Equation 2.11 and Equation 3.7. This yields to

$$R_{\text{beam}} = 4 \frac{U_{\text{sig}} R_w}{U_{\text{bridge}} G} \quad (3.9)$$

with gain G of the amplifier. At the voltage supply, $U_0 = (12.17 \pm 0.01) \text{ V}$ were measured. The JFET is assumed to have a resistance of $R_t = (40 \pm 5) \Omega$, the potentiometers of $R_{t1} = (470 \pm 47) \Omega$. These values are described in section 3.4. The specifications of the parts can be found in Table A.1. For the resistance of the wire in vacuum, we measured a value of $R_{\text{wire}} = (370 \pm 10) \Omega$.

As convectional cooling through hydrogen is neglected in Vassiliev's model, the voltage to calculate the atomic hydrogen flux is gained by subtracting the H_2 voltage from the corresponding H_1 voltage. This way, the cooling effects should be shortened out in an acceptable manner. Then Equation 2.9 with the values from Table 3.2 yields to the fluxes in Table 3.3.

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Constant	Value
Specific resistance (at 20 °C) ρ	$0.092 \Omega \text{ mm}^2 \text{ m}^{-1}$ [16]
Emission coefficient ϵ	0.2 [9]
Temperature coefficient α	$4.7 \times 10^{-3} \text{ K}^{-1}$ [9]
Heat conductivity λ	$174 \text{ W m}^{-1} \text{ K}$ [9]
Wire length L_w	$(50.0 \pm 0.5) \text{ mm}$
Wire diameter d_d	$(5.0 \pm 0.5) \mu\text{m}$ [17]
Room temperature T_0	$(293 \pm 1) \text{ K}$

Table 3.2.: Constants for the calculation of the flux

The obtained fluxes are $2.5 \times 10^{15} \text{ s}^{-1}$. As Schwendler measured the signal without the copper nozzle, his value of 10^{16} s^{-1} is bigger than those with. A loss of H_1 due to recombination on copper is responsible. The flux of the cold beams is about half of the warm ones. The errors are about one order of magnitude bigger than the values due to error propagation. However, the order of magnitude of the fluxes seems to be correct, taking into account the similar magnitude and the relation of warm and cold beam.

z distance	Temperature	Power	Pressure	Flux Φ
1 mm	293 K	40 W	2.5 mbar	$(0.26 \pm 4.37) \times 10^{16} \text{ s}^{-1}$
1 mm	293 K	24 W	3.1 mbar	$(0.26 \pm 4.37) \times 10^{16} \text{ s}^{-1}$
1 mm	4.2 K	24 W	3.1 mbar	$(0.16 \pm 2.36) \times 10^{16} \text{ s}^{-1}$
3 mm	4.2 K	24 W	3.7 mbar	$(0.14 \pm 5.47) \times 10^{16} \text{ s}^{-1}$

Table 3.3.: Calculated H_1 flux for different beams. The pressure is the one in the dissociation area for H_1 . The errors are about order of magnitude bigger than the values due to error propagation.

3.3. Comparison to Former Measurements

In his thesis, Schwendler probed the beam directly emitted by the Teflon hose with a single-wire detector[7]. A comparison of the gathered data can tell about further effects of the nozzle as well as about characteristics of the detector iteration 3.

Especially profiles at room temperature are well comparable and fit those without the nozzle. This means that the nozzle does not affect the beam in its profile. Only the H_1 flux is affected visibly. Schwendler did not account for the cooling by hydrogen by subtracting the H_2 profile. With regard to this, there could have been a higher H_1 flux without the nozzle than he obtained. So, the effect of the nozzle could be even larger. However, the now obtained flux is still smaller. For a flux of magnitude 10^{15} s^{-1} , Schwendler received errors of 10^{16} s^{-1} , which fits the values obtained with revision 3. The obtained divergence of the beam is similar. Schwendler's value is

3. Measurements

within the error of the divergence with nozzle. As it is not specially designed, the shape is sufficient not to expect major differences, similar to the end of the Teflon hose.

3.4. Remark on Errors

The wire is very difficult to handle and cannot be installed into the detector without varying strain. This affects the length of the wire, and thus its total resistance. Only the distance between its attachment screws was measured. An error of 0.5 mm is assumed. Replacing the wire should be avoided as this causes variations. However, when the wire was damaged, it had to be replaced. This is taken into account in the length deviation.

As the detector only measures the resistance change, the total resistance of the wire was measured once in vacuum using a single-wire detector. An absolute error of $10\ \Omega$ is assumed. Some fluctuations caused by vibrations while driving the stage should not be measured since after every driving sequence, a pause of a few seconds was made. A new state of equilibrium of the wire temperature is expected to be reached during this time, too. Opening the chamber can cause changes at the detector position or affect the sensitive wire.

The uncertainties of the detector electronics are known from their specifications. However, the actual resistance of potentiometers and JFETs can only be estimated. Coming from a measured value of $50\ \Omega$ at 6 V trim voltage and the current trim voltage, the error of the JFET is assumed to be $R_t = (40 \pm 5)\ \Omega$. It was attempted to reduce the trimming during the experiment. Between measurements which had to be comparable, no trimming had been performed. The rotation angle of the potentiometers at the bridge led to a resistance of $R_{t1} = (470 \pm 47)\ \Omega$. They have only been slightly manipulated once after being set in the beginning.

By shielding all electronics and wires outside the chamber, the pick-up noise on the signal could be reduced by factor ten. Further fluctuations were taken into account by averaging over multiple data points, taken over one second during measurement. These data points were averaged again. Their standard deviation serves as error on the value.

The bridge itself provides the best values if all resistances are equal. Therefore, the trimming was attempted to be performed so that all arms were as much close to $370\ \Omega$. Because the bridge provides the best results if all resistances are equal, deviations only decrease the signal after the bridge has been balanced. The real flux should always be higher than the calculated one.

During measurements, an increase of the dissociation rate over the first few minutes after ignition could be observed. Until the dissociation stabilised no measurements were performed. Further effects such as pollution or burning of the dissociation area were reduced by a palladium filter for the hydrogen and better cooling.

4. Summary and Outlook

The design and application of a detector for an atomic hydrogen beam has been accomplished. It is small enough to fit in the vacuum chamber and does not severely affect the beam. A thin wire raises its resistance as atomic hydrogen recombines on its surface and deposits energy. A Wheatstone bridge converts the resistance change into a voltage which is amplified by an operational amplifier. This allows the usage of a simple voltmeter for determining the H_1 flux.

This iteration three resolves issues of iteration two, which was not applicable in vacuum. The lack of convectional cooling makes the probe current heat up the wire and unbalances the bridge. This results in saturation of the amplifier. An amplifier with adjustable gain makes the detector more flexible to the flux and different offsets. To balance the bridge inside the vacuum chamber, the effective resistance of two arms can be adjusted with transistors by changing the trim voltage at a voltage divider outside the chamber.

After the functionality has been proven in section 2.3, the new built detector could be used to probe an atomic hydrogen beam.

The beam was probed with and without dissociation (H_1 or H_2). Before the nozzle of a cryostat had been attached, microwave dissociation power and pressure dependency were probed in order to find the optimal values for operation.

Behind the cryostat nozzle at room temperature and cooled down to 4.2 K with helium, the beam was probed for its characteristics. Its profile could be determined to be Gaussian-shaped. The intensity decreases with the distance to the nozzle squared as the beam diverges at an angle of $(16 \pm 5)^\circ$. With the obtained values, the flux was calculated as $2.6 \times 10^{15} \text{ s}^{-1}$. Cooling the beam with liquid helium (4.2 K) reduces the flux by its half. However, the calculated error is about one order of magnitude larger. Nevertheless, this value is of plausible magnitude.

Some characteristics of the cryogenic beam were measured while cooling the cryostat down. Frozen hydrogen on the nozzle reduces the recombination which raises the signal strength, but after one to two hours the nozzle freezes up and the beam can hardly be detected any more. Measurements using the beam should therefore be performed within the first hour of cooling.

The outcomes can be regarded in the further setup of T-REX as they provide information about the beam characteristics. Additionally, the detector or a further iteration will be applied in the further beam run. Probing the beam after the magnetic velocity filter will be the next step.

4. Summary and Outlook

As the beam shall be monitored during the experiment, a further iteration will be built. From the gained experience, some improvements can be made. The wire which is very sensitive and easy to damage will be protected against tears. Some potentiometers which are not accurate enough or sensitive against mechanical effects will be replaced. As the selection of the gain from outside the chamber somehow affected the balancing unit, this feature is planned to be dropped. On the detector itself, the gain shall be still adjustable. Also, the single-wire detector is not needed any more as the design works properly. So, for connections are not needed any more and with another connector, the number of cables can be reduced to two. With a new PCB design, it will be scaled down to halve the size of iteration three. Shielding the detector itself can be applied by casting it in metal loaded synthetic resin. All these improvements will increase the reliability, stability and accuracy while offering the ability to use more than one detector with the same amount of space and connections.

A. Appendix

A.1. Tables

Name	Part	Specification
U_+		+12 V (measured: +12.17 V)
U_-		-12 V
R_b	RN55E7001BRE6	Metal Film Resistor 1/10 W 7 k Ω 0.1 %
R_w	MF1/4DC3700F	Metal Film Resistor 1/4 W 370 Ω 1 %
R_s	MBB02070C1000FCT00	Metal Film Resistor 0.6 W 100 Ω 1 %
R_1	MBB02070C5600FCT00	Metal Film Resistor 0.6 W 560 Ω 1 %
	MBB02070C2000FC100	Metal Film Resistor 0.6 W 200 Ω 1 %
R_3	MBB02070C2000FC100	Metal Film Resistor 0.6 W 200 Ω 1 %
R_4		Metal Film Resistor 300 Ω 1 %
R_{t1} R_{t2}	T73YE501KT20	Trimmer Resistor 500 Ω
R_{t3}	TSM4YJ101KR05	Trimmer Resistor SMD 100 Ω 10 %
R_{G1} R_{G9}		Metal Film Resistor 5.1 k Ω 1 % Metal Film Resistor 390 Ω 1 %
R_{G2} R_{G10}		Metal Film Resistor 470 Ω 1 % Metal Film Resistor 18 Ω 1 % Metal Film Resistor 11 Ω 1 %
R_{G3} R_{G11}		Metal Film Resistor 47 Ω 1 % Metal Film Resistor 2.7 Ω 1 %
R_{G4} R_{G12}		Metal Film Resistor 3.9 Ω 1 % Metal Film Resistor 1 Ω 1 %
C_1	TAP106K016CRW	Tantal Capacitor 16 V 10 μ F 10 %
C_2	TAP104K035SRW	Tantal Capacitor 35 V 0.1 μ F 10 %
JP_1 JP_2		pin header
Wire	W 005200	$l = 5$ cm, $d = 5$ μ m , data sheet[17]
A	LT1168CN8#PBF	Instrumentation Amplifier, data sheet[15]
T	J111	N-channel JFET, data sheet[14]

Table A.1.: List of parts and specifications. Multiple entries in the specification column mark the part to be compounded. The total resistance is marked behind.

A.2. Data Sheets

Part	link
W 005110[17]	http://www.goodfellow.com/G/Wolfram-Draht.html
LT1168CN8#PBF[15]	https://www.mouser.de/datasheet/2/609/1168fa-1267441.pdf
J111[14]	https://www.mouser.de/datasheet/2/308/J111-1120705.pdf

Table A.2.: List of important data sheets

A.3. Figures

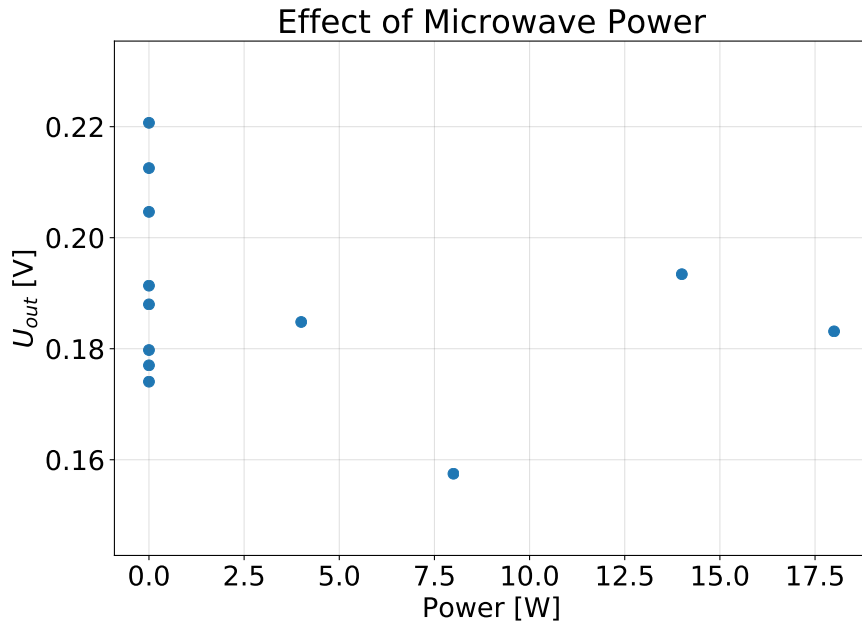


Figure A.1.: For several values of the microwave power but without beam, the signal of the detector was measured to analyse the relation. In the dissociation area, the pressure was 5.9×10^{-2} mbar and in the vacuum chamber 5.2×10^{-6} mbar. There is no visible effect on the measurement from just the microwaves in the dissociation area. The fluctuations are higher than any measurable effect.

A. Appendix

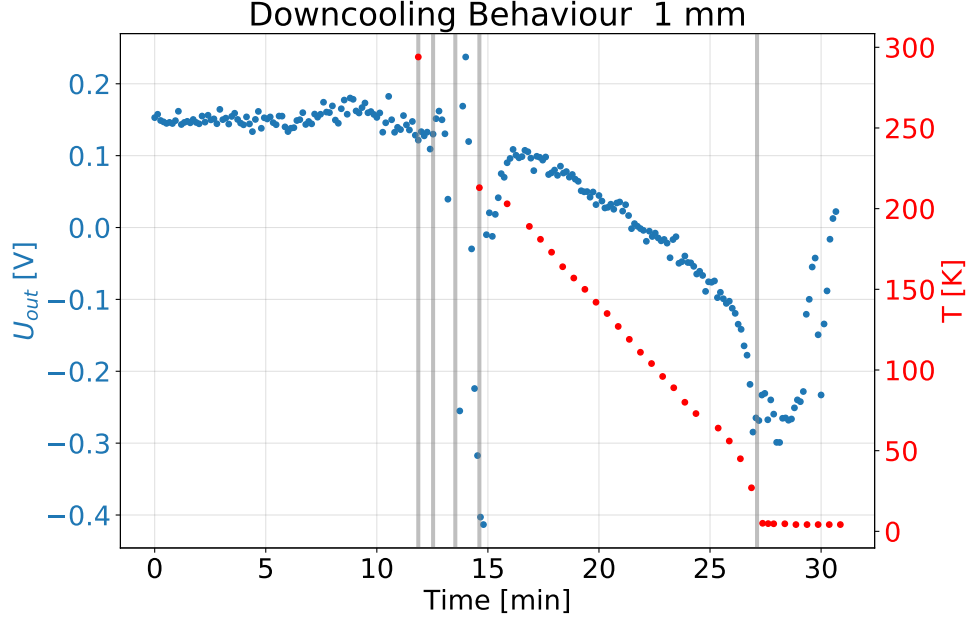


Figure A.2.: Cooling down behaviour of the beam at 1 mm distance to the nozzle. At the first grey marking (from left to right), cooling is applied, the second and third one point out some errors during the H₂ beam was activated. Marking number three shows the ignition of the dissociation with 24 W microwave power and a pressure of 3.8 mbar and gain 10³. With getting colder, the signal decreases until the last marking. It shows the cryostat reaching the temperature $T < 14$ K where hydrogen condenses and freezes at the nozzle. After a short period, the signal increases rapidly.

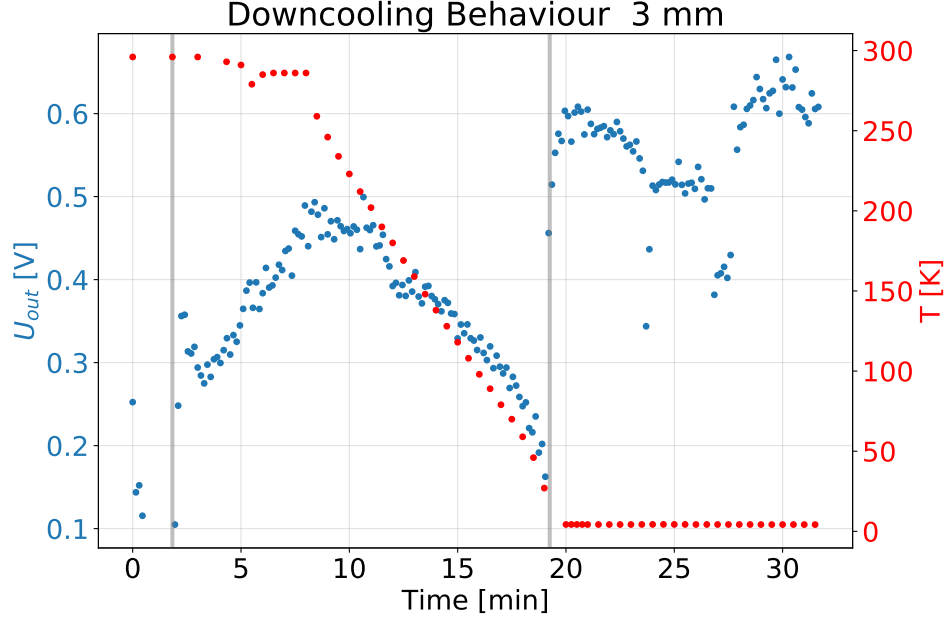


Figure A.3.: Cooling down behaviour of the beam at 3 mm distance to the nozzle. At the first grey marking (from left to right), the dissociation with a power of 25 W is activated. The pressure in the dissociation area is set to 4 mbar. The ascending signal shows an increase of the dissociation efficiency. After around seven minutes cooling is started. With the temperature, the signal decreases. The marker at 19 minutes shows the cryostat reaching a temperature below $T < 14$ K where hydrogen freezes at the nozzle. The coverage of the copper prevents hydrogen from recombining which leads to an increase of the signal. Afterwards, the signal stays high but slowly fluctuates. As the nozzle is affected by frozen hydrogen, the beam could be influenced.

A. Appendix

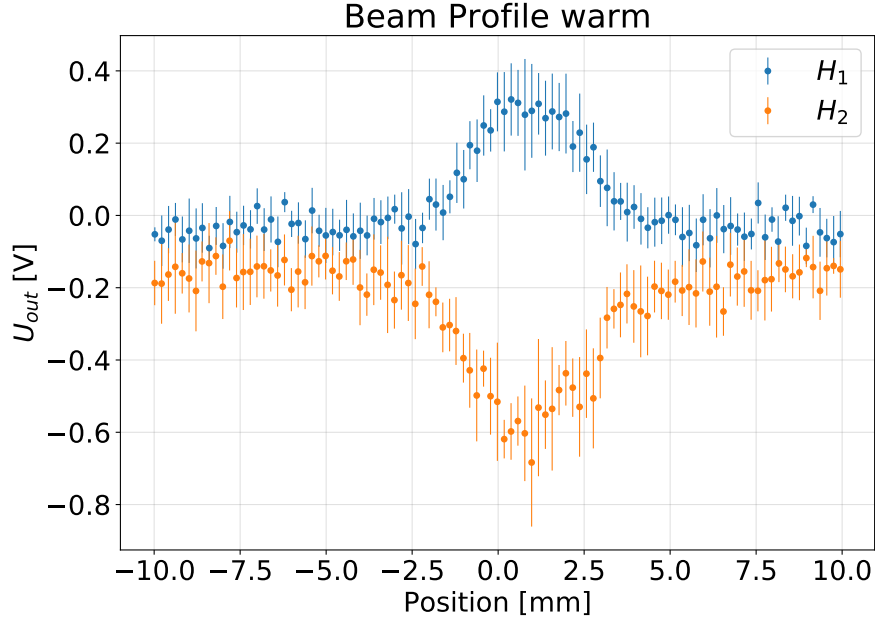


Figure A.4.: Beam profile of H_1 and H_2 at 1 mm distance to the nozzle and 293 K. For H_1 , at a pressure of 3.1 mbar, 24 W microwave power are applied. H_2 is probed at a pressure of 2.6 mbar. The signal is amplified with a gain of 10^3 . For each value, six points are averaged.

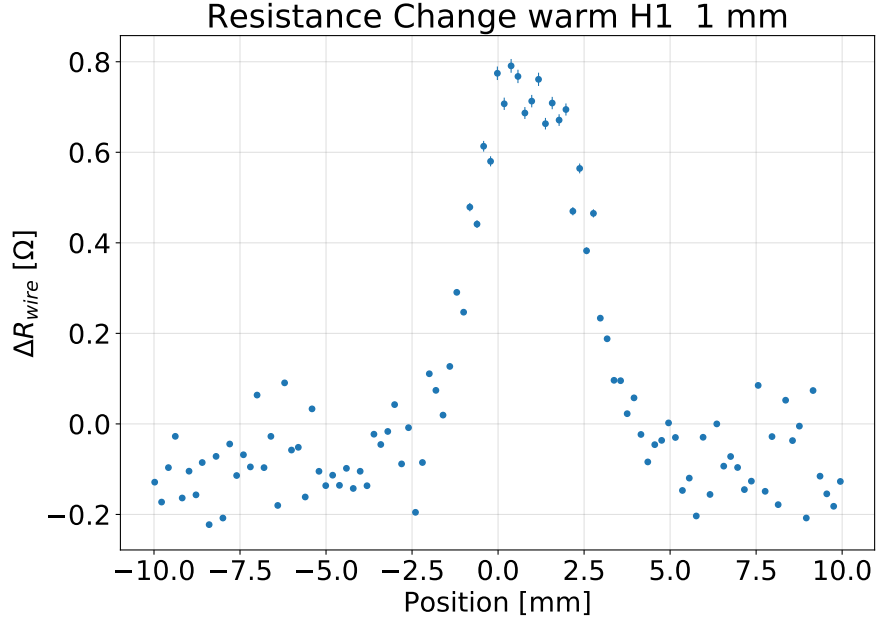


Figure A.5.: Resistance change of H₁ at 1 mm distance to the nozzle and 293 K. At a pressure of 3.1 mbar, 24 W microwave power are applied. The signal is amplified with a gain of 10^3 . The resistance is calculated from the measured voltage as shown in subsection 3.2.4.

A. Appendix

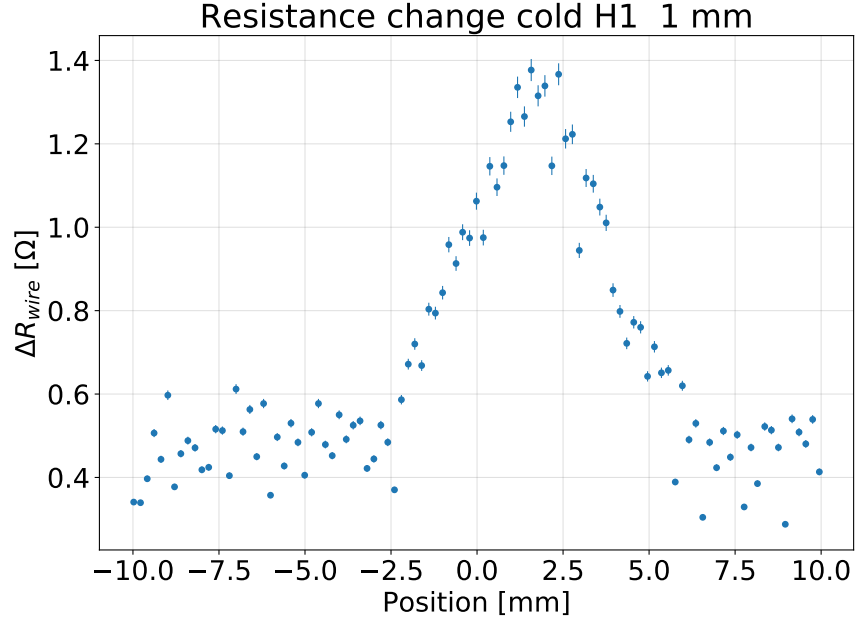


Figure A.6.: Resistance change of H₁ at 1 mm distance to the nozzle and 4.2 K. At a pressure of 3.1 mbar, 24 W microwave power are applied. The signal is amplified with a gain of 10^3 . The resistance is calculated from the measured voltage as shown in subsection 3.2.4.

A. Appendix

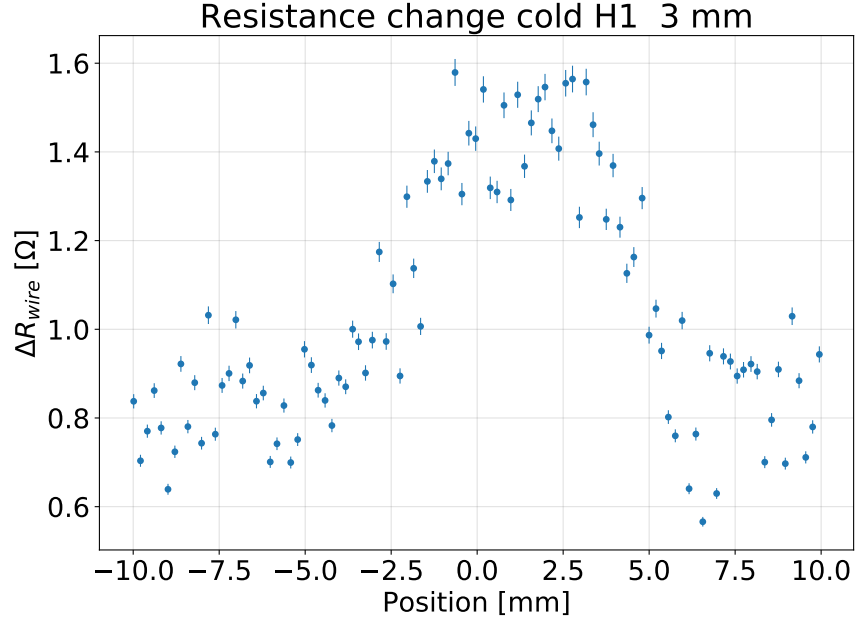


Figure A.7.: Resistance change of H₁ at 3 mm distance to the nozzle and 4.2 K. At a pressure of 3.7 mbar, 24 W microwave power are applied. The signal is amplified with a gain of 10^3 . The resistance is calculated from the measured voltage as shown in subsection 3.2.4.

A.4. Images

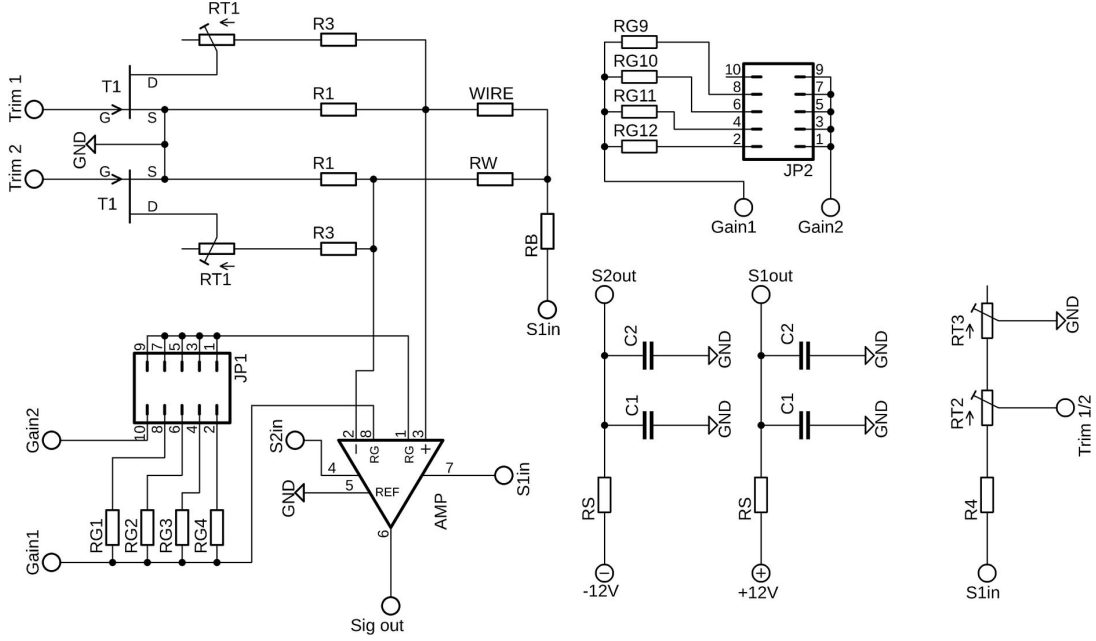


Figure A.8.: Complete circuit diagram of the detector and related electronics. For the sake of simplicity, the power supply and the trimming unit are drawn as individual units. A list of the parts used and their specifications can be found in Table A.1.

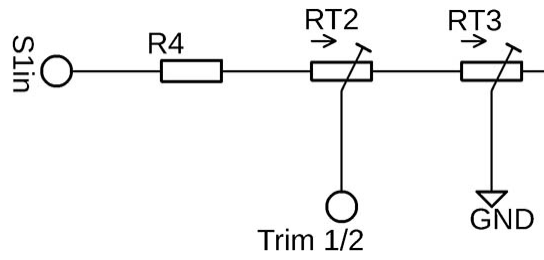


Figure A.9.: Circuit diagram of the outside trimmer. Each JFET gate is controlled by one individual voltage divider unit. $S1_{in}$ is the voltage supply input, R_{T2} and R_{T3} provide a bypass to the transistor gate connected to $Trim_{1/2}$

A. Appendix

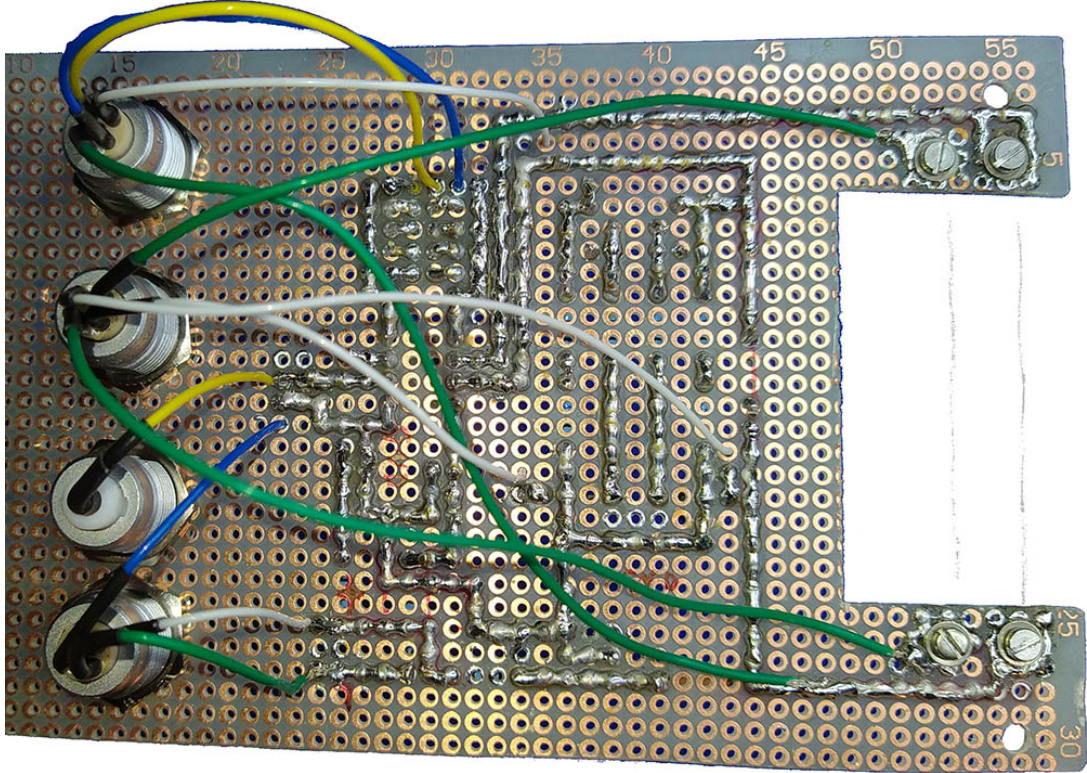


Figure A.10.: Back view of the detector. On the right, two thin wires are recognisable. The right one is part of the Wheatstone detector, the left one of the single-wire detector. The connectors on the left provide the connection to the outside box.

A. Appendix

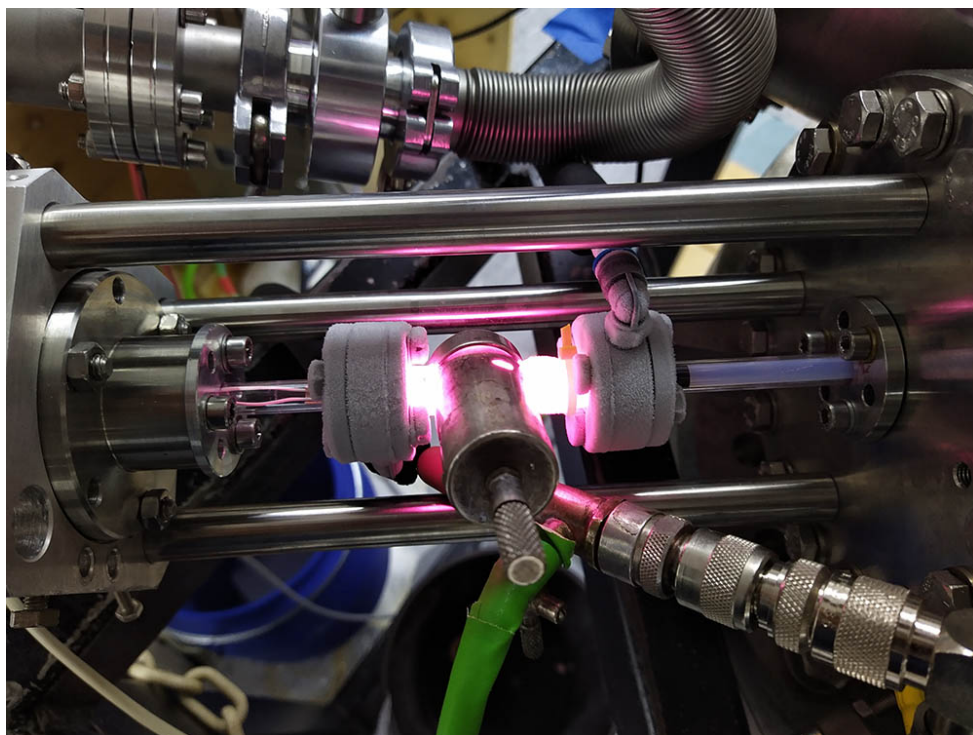


Figure A.11.: When the dissociation is running, the excited hydrogen starts glowing. The cooling is slightly covered with ice. On the right, the Teflon hose is visible through the glass tube.

A. Appendix



Figure A.12.: Iteration 2: previous version of the Wheatstone detector.

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C. German Title, Motivation, Summary and Outlook

Zusätzlich werden hier der deutsche Titel der Arbeit sowie eine deutsche Version von Motivation, Zusammenfassung und Ausblick bereitgestellt.

C.1. Messungen am kryogenen Wasserstoffstrahl

Der Titel der Arbeit lautet auf deutsch "Messungen am kryogenen Wasserstoffstrahl".

C.1.1. Motivation

Ziel dieser Arbeit ist die Entwicklung eines Detektors für atomaren Wasserstoff und die anschließende Charakterisierung eines kryogenen Wasserstoffstrahls. Die Arbeit wird im Rahmen vom Triton-Radius Experiment (T-REX) durchgeführt, dessen Ziel es ist, den Ladungsradius des Tritiumkerns zu bestimmen.

Die an den Detektor gestellten Anforderungen sind die Möglichkeit zur Bestimmung von Strahleigenschaften wie Fluss, Profil und Divergenz bei kompakter Bauweise und der Möglichkeit, die Messungen während des regulären Betriebs, das heißt ohne Beeinflussung des Strahls, vorzunehmen.

Dabei wird die Nutzung des Detektors nicht auf T-REX beschränkt sein, sondern ist für alle Anwendungen möglich, bei denen ein Strahl aus Wasserstoff oder Äquivalentem in einer Vakuumumgebung vermessen werden soll.

Die Idee hinter T-Rex ist die Durchführung von Hochpräzisionslaserspektroskope an Tritium Kernen zur Bestimmung von deren Radius. Die aktuelle Messgenauigkeit von 5 % soll um einen Faktor von 300 verbessert werden, was zu einem besseren Verständnis der Kernkräfte zwischen den drei Nukleonen beitragen könnte.

Dazu werden in einer Mikrowellendissoziierung Tritiummoleküle aufgespalten. Anschließend wird das Gas mittels Helium in einer Düse auf 4.2 K heruntergekühlt. Der hieraus strömende atomare Tritiumstrahl wird in einem magnetischen Geschwindigkeitsfilter purifiziert und der untere Teil des Geschwindigkeitsspektrums in eine magneto-optische Falle (MOT) geleitet. In der MOT werden die T-Atome mittels Lithium weiter heruntergekühlt und anschließend nach Entfernen des Lithiums mittels 2-Photonenspektroskopie vermessen. Da Tritium radioaktiv ist, wird in der Aufbauphase Wasserstoff verwendet, welcher sich als Wasserstoffisotop ausreichend ähnlich verhält.

Der Detektor wird in diesem Experiment hinter der Düse und hinter dem Filter

C. German Title, Motivation, Summary and Outlook

eingesetzt werden, um unter anderem die Qualität der Dissoziierung sowie des Filters zu bestimmen und festzustellen, ob genügend Atome die MOT erreichen. Resultierend können die Funktionalität der Komponenten festgestellt und Verbesserungen am Experimentaufbau vorgenommen werden.

Ein neuer Draht-Detektor wird entwickelt, der gegenüber den gängigen Methoden (Quadrupolmassenspektrometrie (QMS) und Kompressionsrohrmethode) die Vorteile bietet, dass er selektiv gegenüber atomarem Wasserstoff und von kompakter Bauform ist, sowie den Strahl nicht ernstlich beeinflusst.

Nach der Idee von Vassiliev et al. kann ein dünner Draht, der sich durch die Rekombinationsenergie von Wasserstoff erwärmt und somit seinen elektrischen Widerstand erhöht, zum Nachweis von atomarem Wasserstoff verwendet werden. Dieser Widerstand wird mithilfe einer Wheatstonebrücke in eine Spannung umgewandelt, welche dann verstärkt und schließlich mit einem Voltmeter ausgelesen wird. Allerdings traten beim Vorgänger der hier entwickelten Iteration Probleme mit der Balance der Brücke in Vakuum auf, die zur Sättigung der Verstärker führten und damit der Prototyp keine Ergebnisse liefern konnte.

Mithilfe einer externen Einheit zur Balancierung und einem besser auf die Anwendung abgestimmten Verstärker sollen diese Probleme behoben und ein funktionsfähiger Detektor für atomaren Wasserstoff entworfen werden.

Im zweiten Teil der Arbeit sollen Messungen am atomaren Wasserstoffstrahl vorgenommen werden: Die Abhängigkeit des Flusses von Druck und Leistung soll überprüft werden. Nach der Düse soll sowohl mit als auch ohne Heliumkühlung der H_1 -Fluss gemessen werden. Außerdem ist die Charakterisierung des Strahls in den Eigenschaften Profil und Divergenz geplant.

C.1.2. Zusammenfassung und Ausblick

Ein Detektor für atomaren Wasserstoff wurde erfolgreich entwickelt und getestet. Er ist klein genug, um in die Vakuumkammer zu passen und beeinflusst den Strahl nicht nennenswert. Wenn atomarer Wasserstoff auf einen Mikrometerdraht auftrifft und dort rekombiniert, kann die freiwerdende Energie den Draht heizen und somit seinen Widerstand erhöhen. Dieser Widerstand wird mithilfe einer Wheatstonebrücke in eine Spannung umgewandelt. Verstärkt durch einen Operationsverstärker kann diese von einem normalen Voltmeter gemessen werden. Daraus lässt sich der Gesamtfluss an H_1 berechnen.

Mit dieser dritten Iteration werden Probleme des Vorgängermodells gelöst, welche auftraten, wenn der Detektor in einer Vakuumumgebung betrieben wurde. Durch die Verminderung des Drucks sinkt die Konvektionskühlung des Drahts, dessen Widerstand daraufhin ansteigt. Dieses stört das Gleichgewicht in der Brücke und erhöht die zu verstärkende Spannung, was bei Iteration 2 zur Sättigung des Verstärkers führte. Ein Verstärker mit einstellbarer Verstärkungsstufe erhöht die Flexibilität des Detektors gegenüber verschieden starken Flüssen und Offsets. Balanciert werden

C. German Title, Motivation, Summary and Outlook

kann die Brücke von außerhalb der Kammer, indem an zwei Armen der Widerstand je eines JFETs durch Variation einer Steuerspannung geändert wird. Dies ermöglicht eine sehr feine Trimmung.

Nach einigen Funktionstests wurden mit dem neuen Detektor Messungen am atomaren Wasserstoffstrahl durchgeführt.

Bevor die Kryostatdüse angeschlossen wurde, wurde der Strahl auf seine Abhängigkeit von Druck im Dissoziationsbereich und Dissoziationsleistung hin untersucht, um die optimalen Werte für den Betrieb zu finden. Hinter der Düse wurden mit und ohne Wasserstoffdissoziation (H_1 und H_2) sowohl bei Raumtemperatur als auch heliumgekühlt bei 4.2 K Charakteristiken des Strahls untersucht. Dabei konnte ein gaußförmiges Strahlprofil festgestellt werden, welches sich bei kaltem Strahl mehr als Dreieck darstellte. Die Ränder des Strahls wurden durch die langsam mit Wasserstoff zufrierende Düse beeinflusst. Die Strahlintensität sinkt wie erwartet proportional zum Quadrat des Abstands zur Düse und die Auffächerung des Strahls hat einen Winkel von $(16 \pm 5)^\circ$. Mit den erhaltenen Werten lässt sich der Gesamtfluss zu $2.6 \times 10^{15} \text{ s}^{-1}$ berechnen. Der Fluss reduziert sich bei Kühlung um die Hälfte. Obwohl der errechnete Fehler den Fluss um über eine Größenordnung übersteigt, sind die Größenordnung und die Relation der Flüsse mit und ohne Kühlung zueinander, plausibel.

Einige weitere Charakteristiken des kryogenen Strahls konnten beim Herunterkühlen des Kryostaten aufgenommen werden. Gefrorener Wasserstoff bedeckt die Oberfläche der Kupferdüse und verhindert so Rekombination der Wasserstoffatome. Folge ist ein höheres Signal, sobald die Temperatur am Kryostaten den Gefrierpunkt von Wasserstoff erreicht. Während der folgenden ein bis zwei Stunden friert die Düse allerdings so weit zu, dass sie zuerst das Strahlprofil stark beeinflusst und schließlich den Fluss stark vermindert. Messungen am Strahl sollten daher in der ersten Stunde nach dem Abkühlen vorgenommen werden.

Die Ergebnisse können beim weiteren Aufbau von T-REX verwendet werden, um die Strahlführung zu optimieren. Außerdem werden weitere Iterationen des Detektors im weiteren Strahlverlauf zum Einsatz kommen. Der nächste geplante Schritt ist die Untersuchung des Strahls hinter der magnetischen Wasserstofftransportstrecke und deren Charakterisierung.

Da der Strahl während dem Aufbau des Experiments an verschiedenen Stellen überwacht werden soll, wird eine weitere Iteration entwickelt werden. Mithilfe der in dieser Arbeit gewonnenen Erfahrung können einige Verbesserungen gemacht werden: Der sehr leicht abzureißende Draht soll geschützt werden. Weiterhin werden einige wenig akkurate Potentiometer durch feine Trimpotentiometer ersetzt, welche auch weniger anfällig für Stöße sind. Weil die Einstellung der Verstärkungsstufe von außen zu ungewünschtem Verhalten der Balancierungselektronik führte und die Auswahl einer passenden Verstärkungsstufe durch die Ergebnisse dieser Arbeit vereinfacht wurde, wird diese Möglichkeit nicht mehr eingebaut. Die Einstellung soll allerdings am Detektor selbst noch vorgenommen werden können. Der Eindrahtdetektor wird nach

C. German Title, Motivation, Summary and Outlook

dieser Arbeit ebenfalls nicht mehr benötigt. Durch das Wegfallen von vier Anschlüssen und Verwenden anderer Stecker ist es möglich, die Anzahl der Verbindungskabel auf zwei zu reduzieren, wobei eines davon das Signalkabel ist. Ein neues PCB Design kann die Größe auf die Hälfte der dritten Iteration reduzieren. Eingegossen in durch Metall angereichertes Kunstharz soll den Detektor auch in der Kammer gegen Störsignale abschirmen. Diese Verbesserungen werden Zuverlässigkeit, Stabilität und Genauigkeit erhöhen, während sie den Betrieb von doppelt so vielen Detektoren bei gleichem Platz- und Kabelbedarf ermöglichen.